Contents of This Lecture

Learning Objectives of This Lecture

1. Understand the definition of **Laplace transform** and its relationship with Fourier transform;
2. Familiarize with the **basic properties** of Laplace transform.

4.1 Introduction

From Fourier Transform to Laplace Transform

- **Fourier Transform** has a clear physical meaning.
- However, some signals do not satisfy the **Dirichlet conditions** (**sufficient but not necessary conditions for the existence of Fourier Transform**): $\int_{-\infty}^{\infty} |f(t)| \mathrm{d}t < \infty$
- If we multiply by an attenuation factor $e^{-\sigma t}$ (σ>0), then $f(t)e^{-\sigma t}$ converges and and satisfies the Dirichlet conditions.

$u(t)$	$u(t)e^{-\sigma t}$
Growing signal $e^{\alpha t}$ $(\alpha > 0)$	$e^{\alpha t}e^{-\sigma t}$ $(\sigma > \alpha)$
Periodic signal $\cos \omega_1 t$	$e^{-\sigma t} \cos \omega_1 t$

Pierre Simon Laplace (1749-1825)

Laplace Transform (LT)

For a signal $f(t)$, after multiplying by an attenuation factor $e^{-\sigma t}$ (σ>0), it easily satisfies the absolute integrability condition. The new signal's Fourier transform is:

$$F_1(\omega) = F\left[f(t)\cdot e^{-\sigma t}\right] = \int_{-\infty}^{+\infty}\left[f(t)e^{-\sigma t}\right]\cdot e^{-j\omega t}dt$$

$$= \int_{-\infty}^{+\infty} f(t)\cdot e^{-(\sigma+j\omega)t}dt = F(\sigma + j\omega)$$

Complex frequency: $s=\sigma + j\omega$

(Bilateral) Laplace Transform

$$F(s) = \int_{-\infty}^{\infty} f(t)e^{-st}\,dt$$

Unilateral Laplace Transform (practical signals are all **causal signals**. Thus, **"unilateral" often does not appear in the name.)**

$$\boxed{F(s)=\int_{0_-}^{\infty} f(t)e^{-st}dt}$$

$f(t)\xleftrightarrow{LT}F(s)$ $f(t)$--the original function; $F(s)$--the image function.

The unilateral Laplace Transform has included the effect of the initial condition:

$$F(s)=L\left[f(t)\right]=\int_{0_-}^{\infty} f(t)e^{-st}dt$$

Automatically includes the 0_- condition

$$L\left[\frac{df(t)}{dt}\right]=sF(s)-f(0_-)$$

Initial value, equal to 0 if there is no jump

Proof: $\int_{0_-}^{\infty} f'(t)e^{-st}dt = f(t)e^{-st}\Big|_0^{\infty}-\int_0^{\infty} f(t)(e^{-st})'dt$

Integration by parts

$$= f(\infty)e^{-s\infty}-f(0_-)+s\int_0^{\infty} f(t)e^{-st}dt$$

$$= sF(s)-f(0_-)$$

Given the signal $\boldsymbol{x(t) = e^{-\alpha t}u(t)}$， its Laplace transform is ( ).

A $-\dfrac{1}{s+\alpha}$

B $\dfrac{\alpha}{s}$

C $e^{-\alpha s}$

D $\dfrac{1}{s+\alpha}$

提交

Example 4-1: $x(t)=e^{-\alpha t}u(t)$, find its Laplace transform $X(s)$.

Solution: by definition

$$X(s)=\int_{-\infty}^{\infty}x(t)e^{-st}dt=\int_{-\infty}^{\infty}e^{-\alpha t}u(t)e^{-st}dt$$

$$=\int_{0}^{\infty}e^{-\alpha t}e^{-st}dt=\int_{0}^{\infty}e^{-(\alpha+s)t}dt=\int_{0}^{\infty}e^{-(\alpha+\sigma)t}e^{-j\omega t}dt$$

when **Re{α+s}=α+σ>0**, that is $\sigma > -\alpha$, the above integral is integrable

$$X(s)=\frac{1}{-(\alpha+s)}e^{-(\alpha+s)t}\Big|_{0}^{\infty}=\frac{1}{(\alpha+s)},\quad \sigma>-\alpha$$

So,

$$e^{-\alpha t}u(t)\xleftrightarrow{LT}\frac{1}{s+\alpha},\quad \mathrm{Re}\{s\}>-\alpha$$

FT:

$$\begin{cases} F(\omega) = F\left[f(t)\right] = \int_{-\infty}^{\infty} f(t)e^{-j\omega t}dt \\ f(t) = F^{-1}\left[F(\omega)\right] = \dfrac{1}{2\pi}\int_{-\infty}^{+\infty} F(\omega)e^{j\omega t}d\omega \end{cases}$$

The real frequency ω is the **oscillation frequency**.

LT:

$$\begin{cases} F(s) = L\left[f(t)\right] = \int_{0_-}^{\infty} f(t)e^{-st}dt \\ f(t) = L^{-1}\left[F(s)\right] = \dfrac{1}{2\pi j}\int_{\sigma-j\infty}^{\sigma+j\infty} F(s)e^{st}ds \end{cases}$$

The complex frequency $s=\sigma+j\omega$, where ω is the **oscillation frequency** and σ controls the **attenuation speed**.

4.2 Definition of Laplace Transform

Fourier Transform vs. Laplace Transform

Fourier Transform	Laplace Transform
The independent variable ω is a **real number** with a clear physical meaning – **frequency**.	The independent variable $s=\sigma+j\omega$ is a **complex number**, its physical meaning is unclear and it is usually called **complex frequency**.
The Fourier transform of a signal reflects the amplitude and initial phase of different frequency components, i.e., the **spectrum** of the signal.	The Laplace transform of a signal has **no clear physical meaning**.
The Fourier transform of the system's unit impulse response is called the system's **frequency response**, which represents the characteristic that the amplitude and phase of the system output change with the input frequency when cosine signals of different frequencies act on the system.	The Laplace transform of the system's unit impulse response is called the system's **system function**. Although it is abstract, it plays an important role in characterizing system properties and system analysis.
Mainly applied to **frequency analysis** of signals and systems, such as spectrum analysis of modulation, filtering, sampling, etc.	Mainly applied to **solving differential equations**, **analysis of system functions and their zero-pole plots**, etc.

1. Step Function

$$L\left[u(t)\right]=\int_0^{\infty}1\cdot \mathrm{e}^{-st}\mathrm{d}t=\frac{1}{-s}\mathrm{e}^{-st}\Big|_0^{\infty}=\frac{1}{s}$$

2. Single-sided Exponential Function

$$L\left[\mathrm{e}^{-\alpha t}u(t)\right]=\int_0^{\infty}\mathrm{e}^{-\alpha t}\mathrm{e}^{-st}\mathrm{d}t=\frac{\mathrm{e}^{-(\alpha+s)t}}{-(\alpha+s)}\Bigg|_0^{\infty}=\frac{1}{s+\alpha}\quad(\sigma>-\alpha)$$

3. Unit Impulse Signal

$$L\left[\delta(t)\right]=\int_0^{\infty}\delta(t)\cdot\mathrm{e}^{-st}\mathrm{d}t=1$$

$$L\left[\delta(t-t_0)\right]=\int_0^{\infty}\delta(t-t_0)\cdot\mathrm{e}^{-st}\mathrm{d}t=\mathrm{e}^{-st_0}$$

4. $t^n u(t)$

$$L\left[t^n\right]=\int_0^{\infty} t^n \cdot \mathrm{e}^{-st}\mathrm{d}t=\frac{t^n}{-s}\mathrm{e}^{-st}\Big|_0^{\infty}+\frac{n}{s}\int_0^{\infty} t^{n-1}\mathrm{e}^{-st}\mathrm{d}t=\frac{n}{s}\int_0^{\infty} t^{n-1}\mathrm{e}^{-st}\mathrm{d}t$$

so $L\left[t^n\right]=\frac{n}{s}L\left[t^{n-1}\right]$

$n=1 \quad L[t]=\frac{1}{s}L\left[t^0 u(t)\right]=\frac{1}{s}\cdot\frac{1}{s}=\frac{1}{s^2}$

$n=2 \quad L\left[t^2\right]=\frac{2}{s}L[t]=\frac{2}{s}\cdot\frac{1}{s^2}=\frac{2}{s^3}$

$n=3 \quad L\left[t^3\right]=\frac{3}{s}L\left[t^2\right]=\frac{3}{s}\cdot\frac{2}{s^3}=\frac{6}{s^4} \qquad \cdots\cdots$

so $\boxed{L\left[t^n\right]=\frac{n!}{s^{n+1}}}$

Note: The properties refer to those of the unilateral Laplace transform and only apply to causal signals, i.e., signals $f(t)$ that satisfy $f(t)= f(t)u(t)$.

1. Linearity

$$\text{if} \quad f_i(t) \xleftrightarrow{LT} F_i(s)$$

$$\text{then} \quad f(t) = \sum_{i=1}^{N} a_i f_i(t) \xleftrightarrow{LT} F(s) = \sum_{i=1}^{N} a_i F_i(s)$$

Example 4-2: Use the linearity property to find the Laplace transform of $x(t) = e^{-t}u(t) - e^{-2t}u(t)$.

$$\text{Solution: } e^{-t}u(t) \xleftrightarrow{LT} \frac{1}{s+1} \qquad e^{-2t}u(t) \xleftrightarrow{LT} \frac{1}{s+2}$$

$$\text{So} \quad x(t) \xleftrightarrow{LT} \frac{1}{s+1} - \frac{1}{s+2} = \frac{1}{(s+1)(s+2)}$$

Example 4-3: Use the properties of Laplace transform to find the Laplace transform of $\cos(\omega_0 t)u(t)$.

Solution:

$$\cos(\omega_0 t)u(t) = \frac{1}{2}\left(e^{j\omega_0 t} + e^{-j\omega_0 t}\right)u(t)$$

$$e^{j\omega_0 t}u(t) \xleftrightarrow{LT} \frac{1}{s - j\omega_0}$$

$$e^{-j\omega_0 t}u(t) \xleftrightarrow{LT} \frac{1}{s + j\omega_0}$$

$$\therefore \quad \cos(\omega_0 t)u(t) \xleftrightarrow{LT} \frac{s}{s^2 + \omega_0^2}$$

2. Time Delay Property (Time-domain Shift Property)

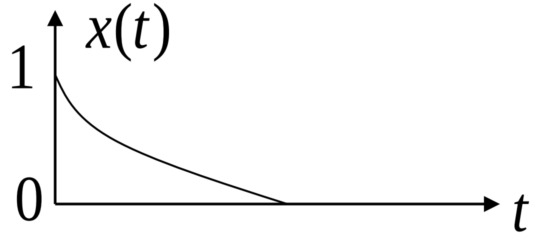
Let $f(t) \xleftrightarrow{LT} F(s)$

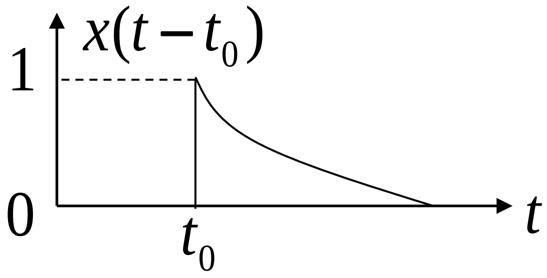
$$f(t-t_0)u(t-t_0) \xleftrightarrow{LT} F(s)e^{-st_0}$$

Note: This property cannot be used if the time-shifted signal is non-causal!

Under the definition of the **unilateral Laplace transform**,

The time delay of $e^{-\alpha t}u(t)$ is $e^{-\alpha(t-t_0)}u(t-t_0)$





单选题 1分

设置

The Laplace transform $F(s)$ of the signal $f(t) = u(t) - u(t-1)$ is（ ）

A $F(s) = \frac{1+e^{-s}}{s}$

B $F(s) = \frac{1-e^{-s}}{s}$

C $F(s) = \frac{1-s}{s}$

D $F(s) = \frac{1-s}{e^{-s}}$

提交

4.4 Basic Properties of Laplace Transform

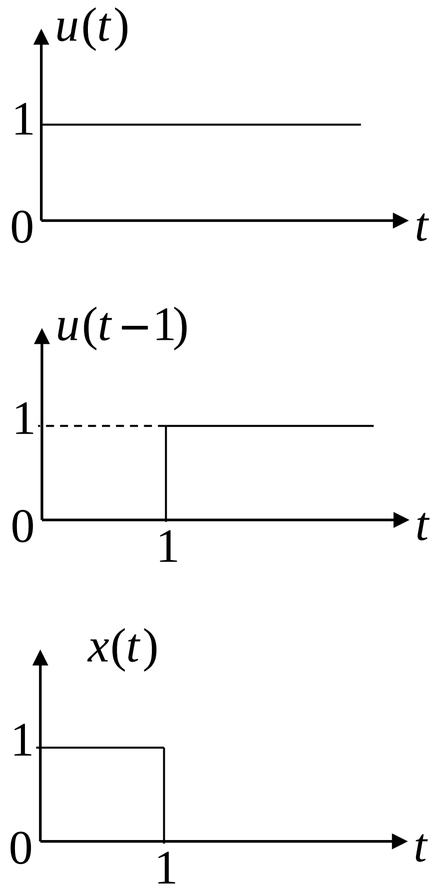
Example 4-4: Find the Laplace transform of $x(t) = u(t) - u(t-1)$.

Solution:

$$u(t) \xleftrightarrow{LT} \frac{1}{s}$$

$$u(t-1) \xleftrightarrow{LT} \frac{1}{s} e^{-s}$$

so $$x(t) = u(t) - u(t-1) \xleftrightarrow{LT} \frac{1-e^{-s}}{s}$$



3. s-domain Shift Property (Frequency Shift Property)

Let $$f(t) \xleftrightarrow{LT} F(s)$$

Then $$f(t)e^{s_0 t} \xleftrightarrow{LT} F(s-s_0)$$

Example 4-5： Use the s-domain shift property to find the Laplace transforms of $e^{-\alpha t}u(t)$ and $te^{-\alpha t}u(t)$.

Solution:

$$u(t) \xleftrightarrow{LT} \frac{1}{s} \qquad tu(t) \xleftrightarrow{LT} \frac{1}{s^2}$$

$$e^{-\alpha t}u(t) \xleftrightarrow{LT} \frac{1}{s+\alpha} \qquad te^{-\alpha t}u(t) \xleftrightarrow{LT} \frac{1}{(s+\alpha)^2}$$

4. Time-domain Scaling Property

if $$f(t) \xleftrightarrow{LT} F(s)$$

then $$f(at) \xleftrightarrow{LT} \frac{1}{a} F\left(\frac{s}{a}\right)$$

The constant *a* > 0， otherwise the Laplace transform is meaningless.

Example 4-6： Use the time-domain scaling property to find the Laplace transform of $e^{-\alpha t}u(t)$.

Solution:

$$e^{-t}u(t) \xleftrightarrow{LT} \frac{1}{s+1}$$

thus, $$e^{-\alpha t}u(t) \xleftrightarrow{LT} \frac{1}{\alpha} \frac{1}{\frac{s}{\alpha}+1} = \frac{1}{\alpha} \frac{\alpha}{s+\alpha} = \frac{1}{s+\alpha}$$

if $$f(t)u(t) \xleftrightarrow{LT} F(s)$$

When there are both time delay and time-domain scaling,

$$f(at-b)u(at-b) \xleftrightarrow{LT} ? \quad (a>0, b>0)$$

First perform time delay, then time-domain scaling:

$$f(t-b)u(t-b) \xleftrightarrow{LT} F\left(s\right)e^{-bs}$$

$$f(at-b)u(at-b) \xleftrightarrow{LT} \frac{1}{a}F\left(\frac{s}{a}\right)e^{-\frac{bs}{a}}$$

It is also OK to perform time-domain scaling first, then time delay.

Example 4-7: Find the Laplace Transform of $\delta(3t-2)$.

Solution 1: First Time Delay, Then Time-domain Compression

From the time-shift property $f(t-t_0)u(t-t_0) \xleftrightarrow{LT} F(s)e^{-st_0}$ and $\delta(t) \xleftrightarrow{LT} 1$

we have: $\delta(t-2) \xleftrightarrow{LT} e^{-2s}$

From the time-domain scaling property $f(at) \xleftrightarrow{LT} \frac{1}{a}F\left(\frac{s}{a}\right)$, we get: $\delta(3t-2) \xleftrightarrow{LT} \frac{1}{3}e^{-\frac{2}{3}s}$

Solution 2: First Time-domain Compression, Then Time Delay

From the time-domain scaling property $f(at) \xleftrightarrow{LT} \frac{1}{a}F\left(\frac{s}{a}\right)$, we get: $\delta(3t) \xleftrightarrow{LT} \frac{1}{3}$

From the time-shift property $f(t-t_0)u(t-t_0) \xleftrightarrow{LT} F(s)e^{-st_0}$:

$$\delta(3t-2) = \delta[3(t-\frac{2}{3})] \xleftrightarrow{LT} \frac{1}{3}e^{-\frac{2}{3}s}$$

4.4 Basic Properties of Laplace Transform

5. Time-domain Differentiation Property

Let $f(t) \xleftrightarrow{LT} F(s)$

$$f'(t) \xleftrightarrow{LT} sF(s) - f(0_-)$$

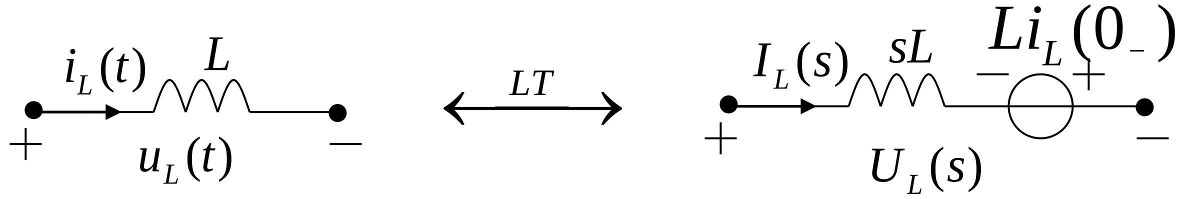
$$f^{(n)}(t) \xleftrightarrow{LT} s^n F(s) - \sum_{i=0}^{n-1} s^{n-i-1} f^{(i)}(0_-)$$

With zero state, differentiating once in the time domain corresponds to multiplying by s in the frequency domain.

Example 4-8: Find the Laplace Transform of inductor voltage $u_L(t)$.

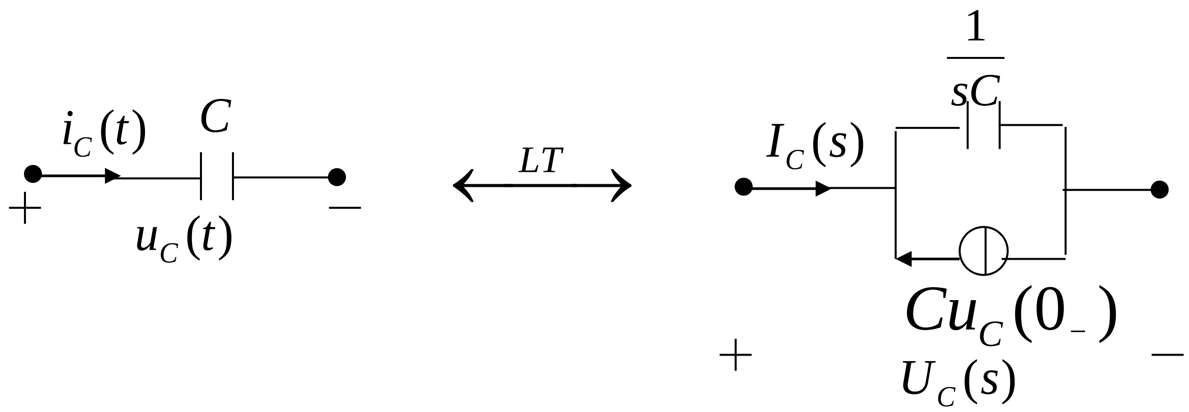
Solution:

$$\boxed{u_L(t) = L\frac{di_L(t)}{dt} \xleftrightarrow{LT} U_L(s) = sLI_L(s) - Li_L(0_-)}$$



Similarly, for **the current in a capacitor**,

$$\boxed{i_C(t) = C\frac{du_C(t)}{dt} \xleftrightarrow{LT} I_C(s) = sCU_C(s) - Cu_C(0_-)}$$



4.4 Basic Properties of Laplace Transform

6. Time-domain Integration Property

$$f(t) \xleftrightarrow{LT} F(s)$$

then $$y(t) = \int_{-\infty}^{t} f(\tau)d\tau = \int_{-\infty}^{0_-} f(\tau)d\tau + \int_{0_-}^{t} f(\tau)d\tau = y(0_-) + \int_{0_-}^{t} f(\tau)d\tau$$

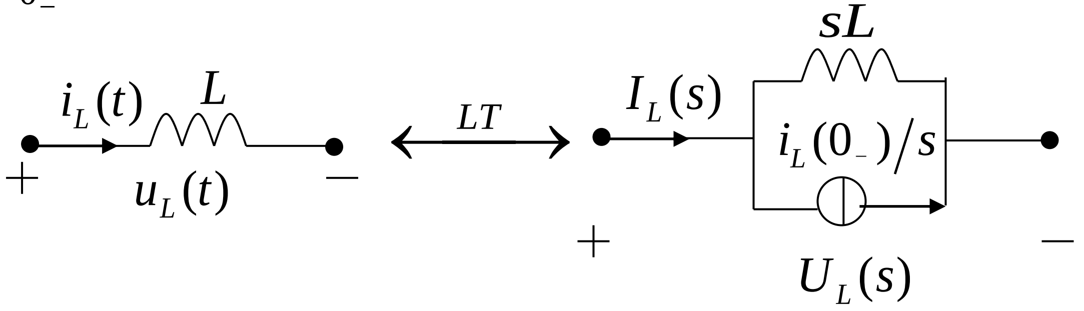
$$y(t) = \int_{-\infty}^{t} f(\tau)d\tau \xleftrightarrow{LT} \frac{y(0_-)}{s} + \frac{F(s)}{s}$$

Example 4-9: Find the Laplace transform of the current $i_L(t)$ in an inductor.

Solution: $$i_L(t) = \frac{1}{L}\int_{-\infty}^{t} u_L(\tau)d\tau = i_L(0_-) + \frac{1}{L}\int_{0_-}^{t} u_L(\tau)d\tau$$

thus,

$$\boxed{i_L(t) \xleftrightarrow{LT} \frac{i_L(0_-)}{s} + \frac{U_L(s)}{sL}}$$



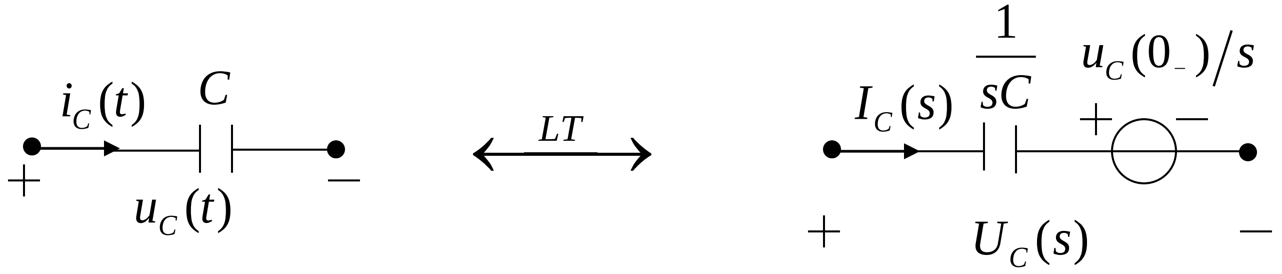
4.4 Basic Properties of Laplace Transform

Similarly, the voltage across a capacitor

$$u_C(t) = \frac{1}{C}\int_{-\infty}^{t} i_C(\tau)d\tau = u_C(0_-) + \frac{1}{C}\int_{0_-}^{t} i_C(\tau)d\tau$$

Its Laplace transform is

$$\boxed{u_C(t) \xleftrightarrow{LT} \frac{u_C(0_-)}{s} + \frac{I_C(s)}{sC}}$$



The **time-domain calculus properties** of Laplace transform are quite convenient when **analyzing the transient state of circuits**.

7. s-domain Differentiation Property

Let $f(t) \xleftrightarrow{LT} F(s)$

s-domain differentiation property:

$$-tf(t) \xleftrightarrow{LT} \frac{dF(s)}{ds} \qquad tf(t) \xleftrightarrow{LT} -\frac{dF(s)}{ds}$$

Example 4-10: Given $e^{-\alpha t}u(t) \xleftrightarrow{LT} \frac{1}{s+\alpha}$, use the s-domain differentiation property to find the Laplace transform of $te^{-\alpha t}u(t)$.

Solution: $\frac{d}{ds}[\frac{1}{s+\alpha}] = \frac{-1}{(s+\alpha)^2}$

Thus, $te^{-\alpha t}u(t) \xleftrightarrow{LT} \frac{1}{(s+\alpha)^2}$

8. s-domain Integration Property

$$\frac{f(t)}{t} \xleftrightarrow{LT} \int_s^{\infty} F(\lambda)d\lambda$$

Example 4-11: Given $(1-e^{-\alpha t})u(t) \xleftrightarrow{LT} \frac{1}{s} - \frac{1}{s+\alpha}$, find the Laplace transform of $\frac{1-e^{-\alpha t}}{t}u(t)$.

Solution:

$$\int_s^{\infty} \left(\frac{1}{\lambda} - \frac{1}{\lambda+\alpha}\right)d\lambda = \ln\lambda\Big|_s^{\infty} - \ln(\lambda+\alpha)\Big|_s^{\infty}$$

$$= -\ln s + \ln(s+\alpha)$$

$$= -\ln\frac{s}{s+\alpha}$$

Thus,

$$\frac{1-e^{-\alpha t}}{t}u(t) \xleftrightarrow{LT} -\ln\frac{s}{s+\alpha}$$

9. Convolution Theorem (Time-domain, Frequency-domain)

Let $$f_1(t) \xleftrightarrow{LT} F_1(s),\ f_2(t) \xleftrightarrow{LT} F_2(s)$$

then $$f_1(t) * f_2(t) \xleftrightarrow{LT} F_1(s) \cdot F_2(s)$$

$$f_1(t) f_2(t) \xleftrightarrow{LT} \frac{1}{2\pi j} F_1(s) * F_2(s)$$

This theorem can be proved using the definition of Laplace transform and the time-delay property.

Example 4-12: Given the unit impulse response and input of a system as follows, find the zero-state response $y(t)$.

$$h(t) = e^{-\alpha t}u(t) \quad , \quad x(t) = u(t)$$

Solution: $$h(t) = e^{-\alpha t}u(t) \xleftrightarrow{LT} \frac{1}{s+\alpha} \qquad x(t) = u(t) \xleftrightarrow{LT} \frac{1}{s}$$

so $$y(t) = x(t) * h(t) \xleftrightarrow{LT} \frac{1}{s(s+\alpha)} = \frac{1}{\alpha}\left(\frac{1}{s} - \frac{1}{s+\alpha}\right)$$

$$y(t) = \frac{1}{\alpha}(1 - e^{-\alpha t})u(t)$$

10. Initial-Value and Final-Value Theorems

Suppose the signal ***f*(*t*)** is **causal**, and the Laplace transforms of ***f*(*t*)** and its derivatives exist, then

(1) Initial-Value Theorem:

$$\lim_{t\to 0_+} f(t) = f(0_+) = \lim_{s\to\infty} SF(s)$$

It represents the high-frequency component (ω→∞) , indicating the mutation of the signal at t=0.

If ***f*(t)** contains an impulse function $k\delta(t)$,

$$F(s) = k + F_1(s),\ \ F_1(s) \text{ is a proper fraction}$$

$$\lim_{t\to 0_+} f(t) = \lim_{s\to\infty}[sF(s) - ks] = \lim_{s\to\infty} sF_1(s)$$

(2) Final-Value Theorem: if ***f***(∞) exists， then

$$\lim_{t\to\infty} f(t) = f(\infty) = \lim_{s\to 0} SF(s)$$

It represents the DC component (ω→0), obtaining the final value of the circuit in steady state.

When circuits and other systems are relatively complex, the initial value and final value of the function can be directly obtained without performing the inverse Laplace transform.

Example 4-13: Given $u(t) \xleftrightarrow{LT} \frac{1}{s}$, use the properties of Laplace transform to find the initial value and final value of $u(t)$ and $\cos(\omega_0 t)u(t)$ respectively.

Solution:

$$u(t)|_{t=0_+} = u(t)|_{t=\infty} = s \cdot \frac{1}{s} = 1$$

$$\cos(\omega_0 t)u(t) \xleftrightarrow{LT} \frac{s}{s^2+\omega_0^2} \qquad \cos(\omega_0 t)|_{t=0_+} = \lim_{s\to\infty} s\frac{s}{s^2+\omega_0^2} = 1$$

$\cos(\omega_0 t)u(t)$ is a constant-amplitude oscillating function (the roots of the denominator of $sF(s)$ are conjugate symmetric on the imaginary axis), so its final value does not exist.

Example 4-14: Use the properties of Laplace transform to find the Laplace transform of $te^{-\alpha t}\cos(\omega_0 t)u(t)$.

Solution: Apply the **Frequency-Shifting Property** and the **s-domain Differentiation Property**.

$$e^{-\alpha t}\cos(\omega_0 t)u(t) \xleftrightarrow{LT} \frac{s+\alpha}{(s+\alpha)^2+\omega_0^2}$$

$$-te^{-\alpha t}\cos(\omega_0 t)u(t) \xleftrightarrow{LT} \frac{d}{ds}\left[\frac{s+\alpha}{(s+\alpha)^2+\omega_0^2}\right]$$

$$\frac{d}{ds}\left[\frac{s+\alpha}{(s+\alpha)^2+\omega_0^2}\right] = \frac{\omega_0^2-(s+\alpha)^2}{[(s+\alpha)^2+\omega_0^2]^2}$$

$$\therefore \quad te^{-\alpha t}\cos(\omega_0 t)u(t) \xleftrightarrow{LT} \frac{(s+\alpha)^2-\omega_0^2}{[(s+\alpha)^2+\omega_0^2]^2}$$

Summary

Property	Time-domain	Complex Frequency Domain
Linearity	$\sum_{i=1}^{N} a_i f_i(t)$	$\sum_{i=1}^{N} a_i F_i(s)$
Time-Delay	$f(t-t_0)u(t-t_0)$	$e^{-st_0}F(s)$
Frequency-Shifting	$f(t)e^{-at}$	$F(s+a)$
Time-Scaling	$f(at) \quad (a>0)$	$\frac{1}{a}F\left(\frac{s}{a}\right)$
Time-Domain Convolution Theorem	$f_1(t)*f_2(t)$	$F_1(s)\cdot F_2(s)$
Frequency-Domain Convolution Theorem	$f_1(t)\cdot f_2(t)$	$F_1(s)*F_2(s)/(2\pi j)$

Note: All the properties apply to **causal signals** only, i.e., the **signals *f*(*t*) that satisfy *f*(*t*)= *f*(*t*)*u*(*t*).**

4.4 Basic Properties of Laplace Transform

Summary

Property	Time-domain	Complex Frequency Domain
Time-domain Differentiation	$\frac{df(t)}{dt}$	$SF(s)-f(0^-)$
Time-domain Integration	$\int_{-\infty}^{t} f(\tau)d\tau$	$\frac{F(s)}{s}+\frac{f^{-1}(0^-)}{s}$
nly. s-domain Differentiation	$-tf(t)$	$\frac{dF(s)}{ds}$
s-domain Integration	$f(t)/t$	$\int_{s}^{\infty} F(\lambda)d\lambda$
Initial Value Theorem	$\lim_{t\to 0^+} f(t)= f(0^+)=\lim_{s\to\infty} sF(s)$ If **$f(t)$** contains an impulse function $\boldsymbol{k\delta(t)}$, $\lim_{t\to 0_+} f(t)=\lim_{s\to\infty} s[F(s)-k]$	
Final Value Theorem	$\lim_{t\to\infty} f(t)= f(\infty)=\lim_{s\to 0} sF(s)$	

Homework

- Repeat the examples in the lecture notes.
- In the textbook:
  - 6.29 (b) $x(t) = tu(t) * \cos(2\pi t)u(t)$

    (e) $x(t) = \int_0^t e^{-3\tau}\cos(2\tau)d\tau$
  - 6.35(a)
  - 6.36(b)

Review of Last Lecture

4.1 Introduction

4.2 Definition of Laplace Transform

4.3 Laplace Transform of Typical Signals

4.4 Basic Properties of Laplace Transform

Laplace Transform (LT)

For a signal $f(t)$, after multiplying by an attenuation factor $e^{-\sigma t}$(where σ is any real number), it easily satisfies the absolute integrability condition. According to the definition of Fourier transform:

$$F_1(\omega) = F\left[f(t)\cdot e^{-\sigma t}\right] = \int_{-\infty}^{+\infty}\left[f(t)e^{-\sigma t}\right]\cdot e^{-j\omega t}dt$$

$$= \int_{-\infty}^{+\infty} f(t)\cdot e^{-(\sigma+j\omega)t}dt = F(\sigma+j\omega)$$

Let $\sigma + j\omega = s$, which has the dimension of frequency and is called complex frequency.

ω is the oscillation frequency, σ controls the rate of attenuation

then $\boxed{F(s) = \int_{-\infty}^{\infty} f(t)e^{-st}\,dt}$

$$f(t) \xleftrightarrow{LT} F(s)$$

Laplace Transform Pair

bilateral Laplace transform

$$\begin{cases} F(s) = L\left[f(t)\right] = \int_{-\infty}^{\infty} f(t) e^{-st} dt \\ f(t) = L^{-1}\left[F(s)\right] = \frac{1}{2\pi j}\int_{\sigma-j\infty}^{\sigma+j\infty} F(s) e^{st} ds \end{cases}$$

Inverse transform

$f(t) \xleftrightarrow{LT} F(s)$ $\quad f(t)$--the original function; $\quad F(s)$--the image function.

Considering that actual signals are all causal signals:

So $\quad F(s) = \int_{0_-}^{\infty} f(t) e^{-st} dt$

unilateral Laplace transform

Because practical systems are **causal**, signals always start acting on the system at a certain moment, and we can regard this moment as t=0; the lower limit of the integral in the above definition is taken as 0_-, and this definition is also valid for impulse signals.

In this chapter and subsequent chapters, unless otherwise specified, it refers to the **unilateral Laplace transform**.

4.2 Definition of Laplace Transform

Fourier Transform vs. Laplace Transform

Fourier Transform	Laplace Transform
The independent variable ω is a **real number** with a clear physical meaning – **frequency**.	The independent variable $s=\sigma+j\omega$ is a **complex number**, its physical meaning is unclear and it is usually called **complex frequency**.
The Fourier transform of a signal reflects the amplitude and initial phase of different frequency components, i.e., the **spectrum** of the signal.	The Laplace transform of a signal has **no clear physical meaning**.
The Fourier transform of the system's unit impulse response is called the system's **frequency response**, which represents the characteristic that the amplitude and phase of the system output change with the input frequency when cosine signals of different frequencies act on the system.	The Laplace transform of the system's unit impulse response is called the system's **system function**. Although it is abstract, it plays an important role in characterizing system properties and system analysis.
Mainly applied to **frequency analysis** of signals and systems, such as spectrum analysis of modulation, filtering, sampling, etc.	Mainly applied to **solving differential equations**, **analysis of system functions and their zero-pole plots**, etc.

Laplace Transforms of Common Signals

Time-domain signal (*t*>0)	S-domain signal
$u(t)$	$\frac{1}{s}$
$e^{-\alpha t}$	$\frac{1}{s+\alpha}$
$\delta(t)$	1
$\delta(t-t_0)$	e^{-st_0}
t^n	$\frac{n!}{s^{n+1}}$

Summary

Property	Time-domain	Complex Frequency Domain
Linearity	$\sum_{i=1}^{N} a_i f_i(t)$	$\sum_{i=1}^{N} a_i F_i(s)$
Time-Delay	$f(t-t_0)u(t-t_0)$	$e^{-st_0} F(s)$
Frequency-Shifting	$f(t)e^{-at}$	$F(s+a)$
Time-Scaling	$f(at) \quad (a>0)$	$\frac{1}{a}F\left(\frac{s}{a}\right)$
Time-Domain Convolution Theorem	$f_1(t) * f_2(t)$	$F_1(s) \cdot F_2(s)$
Frequency-Domain Convolution Theorem	$f_1(t) \cdot f_2(t)$	$F_1(s) * F_2(s) / (2\pi j)$

Note: All the properties apply to **causal signals** only, i.e., the **signals *f*(*t*) that satisfy *f*(*t*)= *f*(*t*)*u*(*t*).**

Summary

Property	Time-domain	Complex Frequency Domain
Time-domain Differentiation	$\dfrac{df(t)}{dt}$	$SF(s)-f(0^-)$
Time-domain Integration	$\int_{-\infty}^{t} f(\tau)d\tau$	$\dfrac{F(s)}{s}+\dfrac{f^{-1}(0^-)}{s}$
s-domain Differentiation	$-tf(t)$	$\dfrac{dF(s)}{ds}$
s-domain Integration	$f(t)/t$	$\int_{s}^{\infty} F(\lambda)d\lambda$
Initial Value Theorem	$\lim_{t\to 0^+} f(t)=f(0^+)=\lim_{s\to\infty} sF(s)$ If **f(t)** contains an impulse function $\boldsymbol{k\delta(t)}$, $\lim_{t\to 0_+} f(t)=\lim_{s\to\infty} s[F(s)-k]$	
Final Value Theorem	$\lim_{t\to\infty} f(t)=f(\infty)=\lim_{s\to 0} sF(s)$	

Find the Laplace transform of $e^{-\alpha t}\sin(\omega_0 t)u(t)$.

A $\dfrac{\omega_0+\alpha}{s^2+\omega_0^2}$

B $\dfrac{s+\alpha}{s^2+\omega_0^2}$

C $\dfrac{\omega_0}{(s+\alpha)^2+\omega_0^2}$

D $\dfrac{s+\alpha}{(s+\alpha)^2+\omega_0^2}$

提交

Find the Laplace transform of $e^{-\alpha t}\sin(\omega_0 t)u(t)$.

Solution: $\sin(\omega_0 t)u(t) = \frac{1}{2j}\left(e^{j\omega_0 t} - e^{-j\omega_0 t}\right)u(t)$

$$e^{j\omega_0 t}u(t) \xleftrightarrow{LT} \frac{1}{s - j\omega_0} \qquad e^{-j\omega_0 t}u(t) \xleftrightarrow{LT} \frac{1}{s + j\omega_0}$$

$$\frac{1}{2j}\left(\frac{1}{s - j\omega_0} - \frac{1}{s + j\omega_0}\right) = \frac{\omega_0}{s^2 + \omega_0^2}$$

$$\therefore \quad \sin(\omega_0 t)u(t) \xleftrightarrow{LT} \frac{\omega_0}{s^2 + \omega_0^2}$$

Using the frequency shift property,

$$e^{-\alpha t}\sin(\omega_0 t)u(t) \xleftrightarrow{LT} \frac{\omega_0}{(s+\alpha)^2 + \omega_0^2}$$

Contents of This Lecture

Learning Objectives of This Lecture

1. Proficiency in finding the **inverse Laplace transform** using the **partial fraction method**;
2. Proficiency in analyzing **circuits** and other **systems** using the **inverse Laplace transform**.

4.5.1 Zeros and Poles of Laplace Transform

$$f(t)=\mathcal{L}^{-1}\{F(s)\}=\frac{1}{2\pi j}\int_{\sigma-j\infty}^{\sigma+j\infty}F(s)e^{st}ds$$

According to the formula of the inverse Laplace transform, contour integration can be applied to solve it. However, in practice, most Laplace transforms are in the form of **rational fractions**:

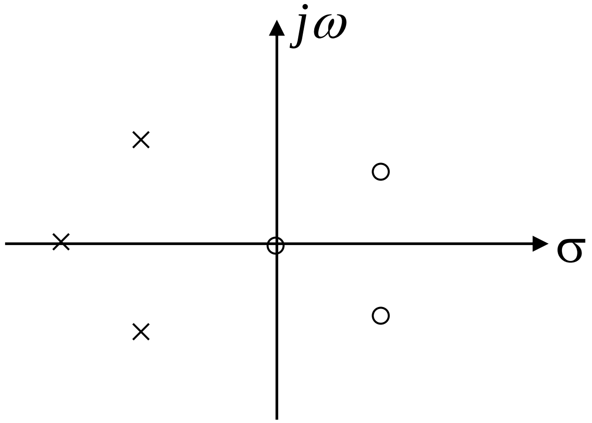
$$F(s)=\frac{N(s)}{D(s)}=\frac{\sum_{i=0}^{m}b_i s^i}{\sum_{i=0}^{n}a_i s^i}$$

Both the numerator and the denominator are **rational polynomials** in ***s*.**

The roots of the numerator polynomial $N(s)=0$ above are called the **zeros** of ***F(s)***; the roots of the denominator polynomial $D(s)=0$ are called the **poles** of ***F(s)***. The above formula can be expressed as:

$$F(s)=\frac{N(s)}{D(s)}=\frac{\prod_{i=1}^{m}(s-z_i)}{\prod_{i=1}^{n}(s-p_i)}$$

In the formula, $\mathbf{z_i}$ and $\boldsymbol{p_i}$ are the zeros and poles of ***F(s)***, respectively. The zeros and poles can be marked on the s-plane as shown in the figure.



4.5.2 Finding the Inverse Laplace Transform

When analyzing systems using the Laplace transform method, the final step is to perform the **inverse Laplace transform** on the image function.

1. Partial Fraction Method (Heaviside Method)

(1) The Case of Simple Real Poles? $(p_1 \cdots p_n)$

When $m < n$, $F(s) \overset{Decomposition}{==} \dfrac{k_1}{s-p_1} + \cdots + \dfrac{k_n}{s-p_n}$ **(**Proper Fraction**)**

where $k_i = (s-p_i)F(s)\big|_{s=p_i}$ (Residue) (See the textbook for the proof)

It is known from the properties of Laplace transform：

$$Ae^{-\alpha t}u(t) \overset{LT}{\longleftrightarrow} \frac{A}{s+\alpha}$$

$$\boxed{F(s) \xrightarrow{L^{-1}} f(t) = \left(k_1 e^{p_1 t} + \cdots + k_n e^{p_n t}\right)u(t)}$$

单选题 1分

设置

Given $F(s)=\frac{s+3}{s^2+3s+2}$, then the Inverse Laplace Transform of $f(t)$ is ( ).

A $f(t)=(2e^{-t}+e^{-2t})u(t)$

B $f(t)=(2e^{t}-e^{2t})u(t)$

C $f(t)=(2e^{-t}-e^{-2t})u(t)$

D $f(t)=(e^{-t}+2e^{-2t})u(t)$

提交

Example 4-15: $F(s)=\dfrac{s+3}{s^2+3s+2}$, find its Inverse Laplace Transform ***f*(*t*)**.

Solution: $F(s)=\dfrac{s+3}{(s+1)(s+2)}=\dfrac{k_1}{(s+1)}+\dfrac{k_2}{(s+2)}$

$$k_1=(s+1)F(s)\big|_{s=-1}=\left.\frac{s+3}{s+2}\right|_{s=-1}=2$$

$$k_2=(s+2)F(s)\big|_{s=-2}=\left.\frac{s+3}{s+1}\right|_{s=-2}=-1$$

So: $F(s)=\dfrac{2}{(s+1)}-\dfrac{1}{(s+2)}$

Its Inverse Laplace Transform: $f(t)=(2e^{-t}-e^{-2t})u(t)$

Even if ***F*(*s*)** is not a proper fraction, i.e. ***n*<*m*,** it can also be expressed as the sum of a polynomial in ***s*** and a **proper fraction**:

Example:

$$F(s)=\frac{s^3+4s^2+6s+5}{s^2+3s+2}$$

Perform polynomial long division to extract the high-degree term:

$$\begin{array}{r|l} & s+1 \\ \hline s^2+3s+2 & s^3+4s^2+6s+5 \\ & s^3+3s^2+2s \\ \hline & s^2+4s+5 \\ & s^2+3s+2 \\ \hline & s+3 \end{array}$$

$$F(s)=\frac{s^3+4s^2+6s+5}{s^2+3s+2}=s+1+\frac{s+3}{s^2+3s+2}$$

The inverse Laplace transform of ***s*+1** in the above equation is:

$$\delta'(t)+\delta(t)$$

For the inverse transform of a **rational proper fraction** in Laplace transform, solve it using the **partial fraction method**.

(2) The case where poles include conjugate complex roots $(p_{1,2} = -\alpha \pm j\beta)$

$$F(s) = \frac{A(s)}{D(s)\left[(s+\alpha)^2 + \beta^2\right]}$$

$$= \frac{A(s)}{D(s)(s+\alpha-j\beta)(s+\alpha+j\beta)}$$

where $D(s)$ is the remaining part of the denominator after removing the conjugate complex roots.

Set $F_1(s) = \dfrac{A(s)}{D(s)}$

then $F(s) = \dfrac{F_1(s)}{(s+\alpha-j\beta)(s+\alpha+j\beta)}$

$$\overset{Decomposition}{=\!=} \frac{K_1}{s+\alpha-j\beta} + \frac{K_2}{s+\alpha+j\beta} + \ldots$$

where $K_{1,2}=\dfrac{F_1(-\alpha\pm j\beta)}{\pm 2j\beta}$ (residue)

K_1、K_2 have a conjugate relationship: $K_{1,2}=A\pm jB$

Thus, the inverse transform of the part related to the conjugate complex roots is $\xrightarrow{\mathrm{LT}^{-1}}$

$$f_c(t)=e^{-\alpha t}(K_1e^{j\beta t}+K_1^*e^{-j\beta t})u(t)$$
$$=e^{-\alpha t}\{(A+jB)[\cos(\beta t)+j\sin(\beta t)]+(A-jB)[\cos(\beta t)-j\sin(\beta t)]\}u(t)$$
$$=2e^{-\alpha t}\left[A\cos\left(\beta t\right)-B\sin\left(\beta t\right)\right]u(t)$$

α--attenuation factor； β--oscillation frequency

Example 4-16: $F(s)=\dfrac{s^2+3}{(s+2)(s^2+2s+5)}$, find its Inverse Laplace Transform ***f*(*t*)**.

Solution:

Let $F(s)=\dfrac{K_1}{s+2}+\dfrac{K_2}{s+1-j2}+\dfrac{K_2^*}{s+1+j2}$

$\alpha=1,$

$\beta=2,$ let $\beta>0$

$$K_1=(s+2)F(s)|_{s=-2}=\left.\frac{s^2+3}{s^2+2s+5}\right|_{s=-2}=\frac{7}{5}$$

$$K_2=(s+1-j2)F(s)\Big|_{s=-1+j2}=\left.\frac{s^2+3}{(s+2)(s+1+j2)}\right|_{s=-1+j2}=-\frac{1}{5}+j\frac{2}{5}$$

$A=-\dfrac{1}{5},B=\dfrac{2}{5}$

The time-domain signal corresponding to the conjugate complex roots is:

$$f_c(t)=2e^{-\alpha t}\left[A\cos\left(\beta t\right)-B\sin\left(\beta t\right)\right]u(t)$$

$$f(t)=\left[\frac{7}{5}e^{-2t}-\frac{2}{5}e^{-t}(\cos 2t+2\sin 2t)\right]u(t)$$

Example 4-17: $F(s)=\dfrac{s+\gamma}{(s+\alpha)^2+\beta^2}$, find its Inverse Laplace Transform ***f(t)***.

Solution: This function **only has conjugate complex roots**, so there is **no need to use the partial fraction method**.

$$\text{known}\quad \mathrm{LT}[\cos(\beta t)u(t)]=\frac{s}{s^2+\beta^2},\quad \mathrm{LT}[\sin(\beta t)u(t)]=\frac{\beta}{s^2+\beta^2}$$

$$\mathrm{LT}[e^{-\alpha t}\cos(\beta t)u(t)]=\frac{s+\alpha}{(s+\alpha)^2+\beta^2},\quad \mathrm{LT}[e^{-\alpha t}\sin(\beta t)u(t)]=\frac{\beta}{(s+\alpha)^2+\beta^2}$$

$$F(s)=\frac{s+\alpha+\gamma-\alpha}{(s+\alpha)^2+\beta^2}=\frac{s+\alpha}{(s+\alpha)^2+\beta^2}+\frac{(\gamma-\alpha)}{\beta}\cdot\frac{\beta}{(s+\alpha)^2+\beta^2}$$

$$f(t)=e^{-\alpha t}\left[\cos(\beta t)+\frac{\gamma-\alpha}{\beta}\sin(\beta t)\right]u(t)$$

(3) The case where poles include multiple roots(a k-fold p_1, simple roots $p_2 \ldots p_n$)

$$F(s) = \frac{A(s)}{D(s)} = \frac{A(s)}{(s-p_1)^k D_1(s)}$$

$$= \frac{K_{11}}{(s-p_1)^k} + \frac{K_{12}}{(s-p_1)^{k-1}} + \cdots + \frac{K_{1k}}{(s-p_1)} + \sum_{i=2}^{n} \frac{K_i}{s-p_i}$$

$$= \sum_{i=1}^{k} \frac{K_{1i}}{(s-p_1)^{k-i+1}} + \sum_{i=2}^{n} \frac{K_i}{s-p_i}$$

To find K_i and K_{11}, the method is the same as the first case:

$$K_{11} = (s-p_1)^k F(s)\Big|_{s=p_1}$$

To find other coefficients,

Let $F_1(s) = (s-p_1)^k \cdot F(s)$

$$\text{when } i=2, \quad K_{12} = \frac{\mathrm{d}}{\mathrm{d}s} F_1(s)\Big|_{s=p_1}$$

$$\text{when } i=3, \quad K_{13} = \frac{1}{2}\frac{\mathrm{d}^2}{\mathrm{d}s^2} F_1(s)\Big|_{s=p_1}$$

$$K_{1i} = \frac{1}{(i-1)!}\frac{\mathrm{d}^{i-1}}{\mathrm{d}\,s^{i-1}}F_1(s)\bigg|_{s=p_1} \qquad (i=1,2,\cdots k)$$

$$F_1(s) = (s-p_1)^k \cdot F(s)$$

The inverse transform of the multiple root part $\xrightarrow{\mathrm{LT}^{-1}}$:

$$f_c(t) = e^{p_1 t}\left[\frac{K_{11}}{(k-1)!}t^{k-1} + \frac{K_{12}}{(k-2)!}t^{k-2}\cdots + \frac{K_{1i}}{(k-i)!}t^{k-i}\cdots + K_{1k}\right]u(t)$$

Example 4-18: $F(s)=\dfrac{s-2}{s(s+1)^3}$, find its Inverse Laplace Transform ***f*(*t*)**.

Solution: Let

$$F(s)=\frac{K_{11}}{(s+1)^3}+\frac{K_{12}}{(s+1)^2}+\frac{K_{13}}{(s+1)}+\frac{K_2}{s}$$

First, find the coefficient K_2 corresponding to the simple root 0:

$$K_2=sF(s)\big|_{s=0}=\left.\frac{s-2}{(s+1)^3}\right|_{s=0}=-2$$

Let $F_1(s)=(s+1)^3F(s)=\dfrac{s-2}{s}$

$$K_{11} = F_1(s)|_{s=-1} = \left.\frac{s-2}{s}\right|_{s=-1} = 3$$

$$K_{12} = \frac{1}{(3-2)!}\left.\frac{dF_1(s)}{ds}\right|_{s=-1} = \frac{d}{ds}\left.\left[\frac{s-2}{s}\right]\right|_{s=-1} = \left.\frac{2}{s^2}\right|_{s=-1} = 2$$

$$K_{13} = \frac{1}{(3-1)!}\left.\frac{d^2F_1(s)}{ds^2}\right|_{s=-1} = \frac{1}{2}\cdot\left.\frac{-4}{s^3}\right|_{s=-1} = 2$$

$$f(t) = \left[e^{-t}\left(\frac{K_{11}}{2!}t^2 + K_{12}t + K_{13}\right) + K_2\right]u(t)$$

$$= \left[e^{-t}\left(\frac{3}{2}t^2 + 2t + 2\right) - 2\right]u(t)$$

Special Case: Irrational Expression Containing $e^{-\alpha s}$

The term $e^{-\alpha s}$ does not participate in the partial fraction operation, and the time – shifting property is used when solving.

Example 4-19: Find the Inverse Laplace Transform of $\dfrac{e^{-2s}}{s^2+3s+2}$.

Solution: $$\frac{e^{-2s}}{s^2+3s+2} = F_1(s)e^{-2s}$$

$$F_1(s) = \frac{1}{s+1} + \frac{-1}{s+2}$$

So $f_1(t) = L^{-1}\left[F_1(s)\right] = \left(e^{-t} - e^{-2t}\right)u(t)$

So $f\left(t\right) = f_1\left(t-2\right) = \left[e^{-(t-2)} - e^{-2(t-2)}\right]u(t-2)$

4.6.1 Solving Differential Equations Using Laplace Transform

Steps for Solving Differential Equations Using Laplace Transform:

(1) Take the Laplace transform of the differential equation, paying attention to applying the differentiation property;

(2) Solve the algebraic equation in the s-domain to find the Laplace transform of the response;

(3) Find the inverse Laplace transform, i.e., find the time-domain expression of the response.

Example 4-20: Given

$$\frac{d^2y(t)}{dt^2}+3\frac{dy(t)}{dt}+2y(t)=\frac{dx(t)}{dt}+4x(t) \qquad y'(0^-)=y(0^-)=1 \qquad x(t)=e^{-3t}u(t)$$

Use the s-domain analysis method to find the zero-state response $y_{zs}(t)$.

Solution:

Take the Laplace transform of both sides of the equation, without initial conditions.

$$\frac{d^2y(t)}{dt^2}+3\frac{dy(t)}{dt}+2y(t)=\frac{dx(t)}{dt}+4x(t)$$

$$s^2Y_{zs}(s)+3sY_{zs}(s)+2Y_{zs}(s)=sX(s)+4X(s)$$

Rearrange the above equation to find the Laplace transform of the zero-state response.

$$(s^2+3s+2)Y_{zs}(s)=(s+4)X(s)$$

So, $Y_{zs}(s)=\dfrac{(s+4)}{(s^2+3s+2)}X(s)=H(s)X(s)$

Since $$X(s)=\frac{1}{s+3}$$

Obtain the Laplace transform of the zero-state response:

$$Y_{zs}(s)=\frac{(s+4)}{(s^2+3s+2)(s+3)}=\frac{(s+4)}{(s+1)(s+2)(s+3)}$$

$$Y_{zs}(s)=\frac{\frac{3}{2}}{s+1}-\frac{2}{s+2}+\frac{\frac{1}{2}}{s+3}$$

Find the inverse Laplace transform to obtain the **zero-state response** of the system:

$$y_{zs}(t)=(\frac{3}{2}e^{-t}-2e^{-2t}+\frac{1}{2}e^{-3t})u(t)$$

Try to find the zero-state response using the time-domain analysis method in Chapter 2 and compare the results.

4.6.2 Solving Circuits Using Laplace Transform

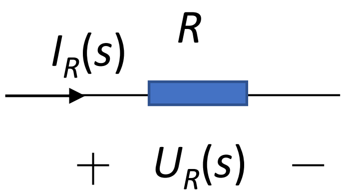
$s-$domain Component Models:

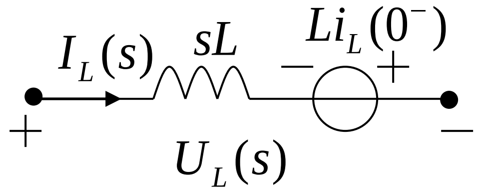
1) Loop Analysis (KVL)

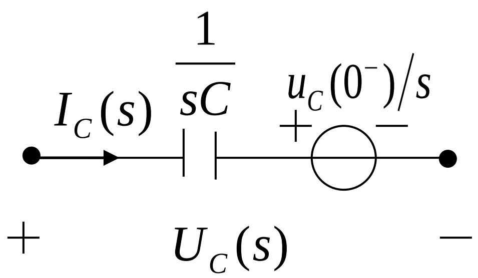
$$U_R(s) = RI_R(s)$$

$$U_L(s) = sLI_L(s) - Li_L(0^-)$$

$$U_C(s) = \frac{1}{sC} I_C(s) + \frac{1}{s} u_c(0^-)$$





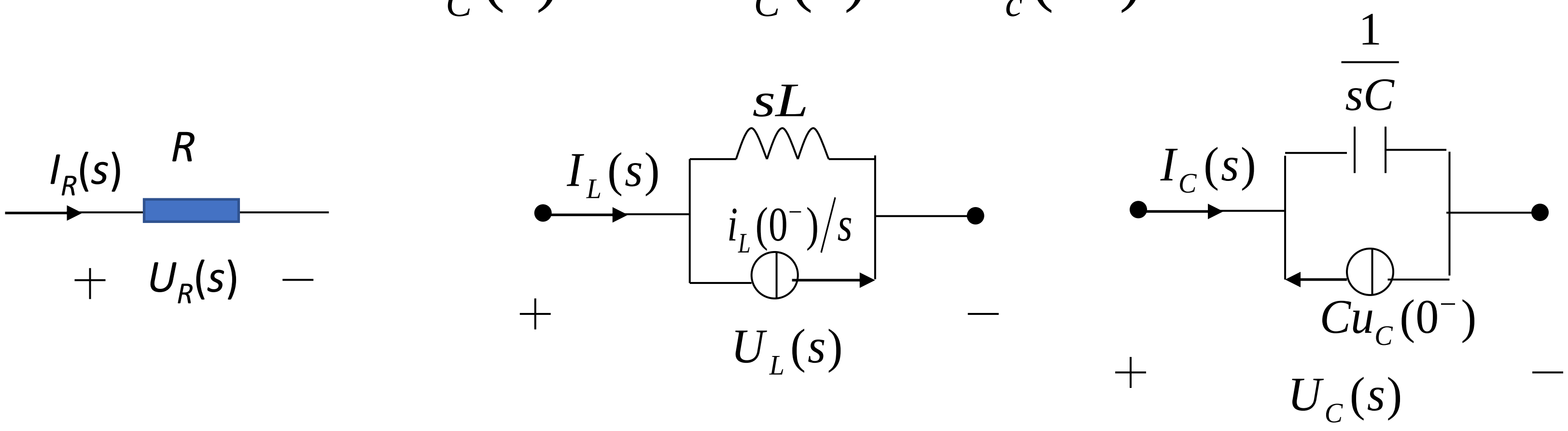


2) Node Analysis (KCL)

$$I_R = \frac{1}{R} U_R(s)$$

$$I_L(s) = \frac{1}{sL} U_L(s) + \frac{1}{s} i_L(0^-)$$

$$I_C(s) = sCU_C(s) - Cu_c(0^-)$$



s-domain Circuit Analysis Method

1. Replace each component in the network with its **s-domain model;**
Write the signal source as a transformed expression ($\frac{E}{s}$ or $\frac{E}{sR}$, $U_s(s)$ or $I_s(s)$)

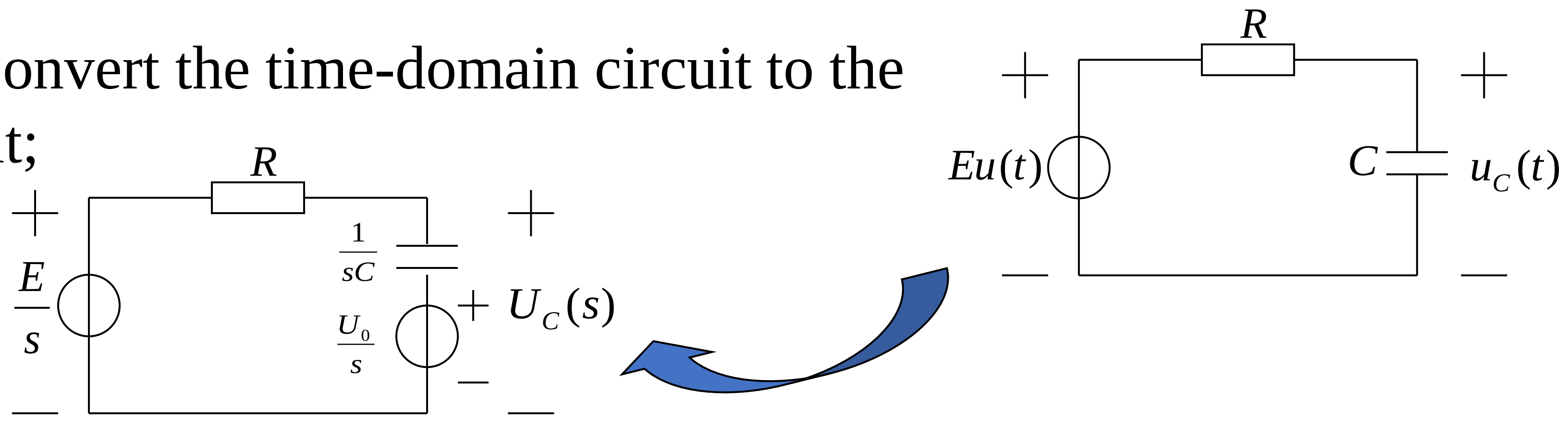
2. Analyze the constructed s-domain model diagram using KVL and KCL to obtain the required transformed system equations.
(The numerical operations performed are algebraic relationships, similar to resistive networks; Thevenin's theorem and Norton's theorem are both applicable; It is suitable for network analysis with many nodes or loops.)

3. Find the **Laplace transform** of the response.

4. Find the **inverse Laplace transform**, i.e., find the time-domain expression of the response.

Example 4-21: An RC circuit is shown in the figure. Given the initial voltage on the capacitor $\boldsymbol{u_C(0^-)=U_0}$, try to find the capacitor voltage $\boldsymbol{u_C(t)}$ for $\boldsymbol{t>0}$.

Solution: (1) Convert the time-domain circuit to the s-domain circuit;



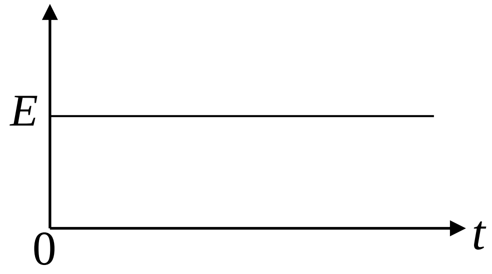
(2)List the circuit equation in the s-domain and find the Laplace transform of the output;

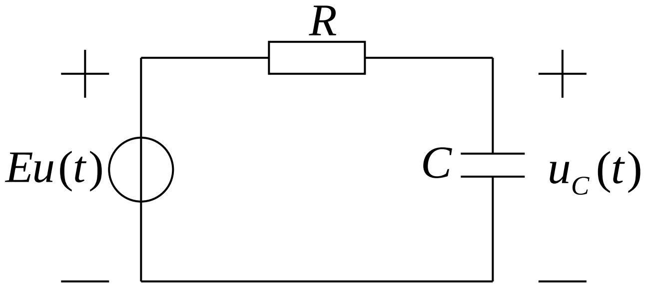
The Laplace transform of the capacitor voltage can be directly written as:

$$U_C(s)=\frac{\frac{1}{sC}}{R+\frac{1}{sC}}\frac{E-U_0}{s}+\frac{U_0}{s}=\frac{1}{sRC+1}\frac{E-U_0}{s}+\frac{U_0}{s}$$

(3)Find the time-domain expression of the response by inverse Laplace transform.

$$U_C(s)=\frac{\frac{1}{RC}(E-U_0)}{s(s+\frac{1}{RC})}+\frac{U_0}{s}=\frac{E-U_0}{s}-\frac{(E-U_0)}{s+\frac{1}{RC}}+\frac{U_0}{s}=\frac{E}{s}-\frac{(E-U_0)}{s+\frac{1}{RC}}$$



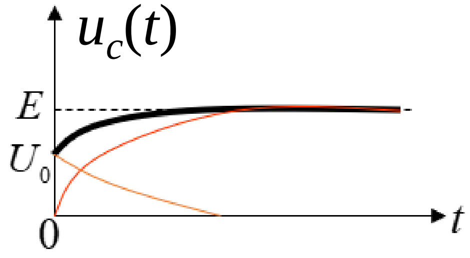


The voltage across the capacitor is obtained.

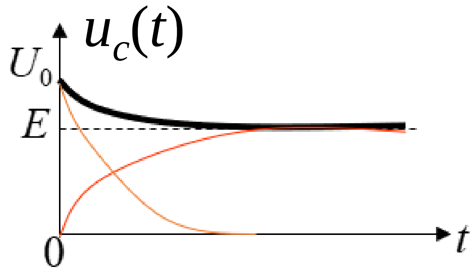
$$u_C(t)=[E-(E-U_0)e^{-\frac{t}{RC}}]u(t)$$ (black wire)

$$u_{Czs}(t)=E(1-e^{-\frac{t}{RC}})u(t)$$ (red wire)

$$u_{Czi}(t)=U_0e^{-\frac{t}{RC}}u(t)$$ (orange wire)

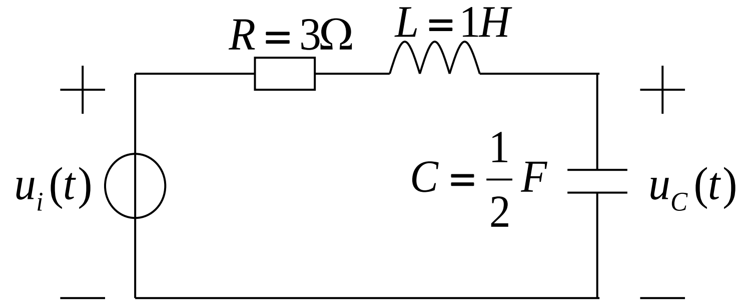


$E>U_0$



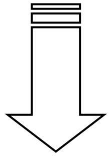
$E<U_0$

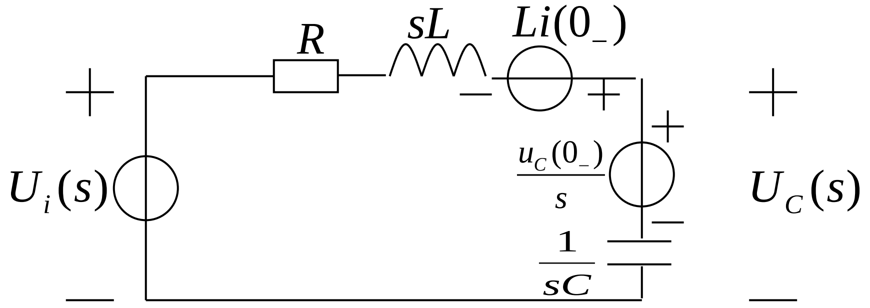
Example 4-22: An RLC series circuit is shown in the figure. Given the initial voltage on the capacitor $\boldsymbol{u_C(0_-)=1\text{ V}}$, the initial current in the loop $\boldsymbol{i(0_-)=1\text{ A}}$, and the input voltage $\boldsymbol{u_i(t)=e^{-3t}u(t)}$. Try to find the capacitor voltage $\boldsymbol{u_C(t)}$ for $\boldsymbol{t>0}$.



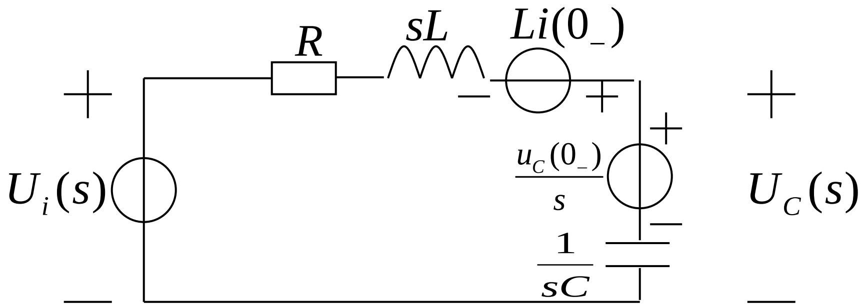
Solution:

(1) Convert the time-domain circuit to the s-domain circuit.





(2) List the circuit equation in the s-domain and find the Laplace transform of the response.



$$U_C(s)=\frac{\frac{1}{sC}}{R+sL+\frac{1}{sC}}\left[U_i(s)-\frac{u_C(0_-)}{s}+Li(0_-)\right]+\frac{u_C(0_-)}{s}$$

$$=\frac{\frac{2}{s}}{3+s+\frac{2}{s}}\left[U_i(s)-\frac{1}{s}+1\right]+\frac{1}{s}$$

$$=\frac{2U_i(s)}{s^2+3s+2}-\frac{2}{s^2+3s+2}\left(\frac{1-s}{s}\right)+\frac{1}{s}$$

$$U_{Czs}(s)=\frac{2}{s^2+3s+2}U_i(s)=\frac{2}{s^2+3s+2}\cdot\frac{1}{s+3}$$

$$=\frac{2}{(s+1)(s+2)(s+3)}=\frac{1}{s+1}-\frac{2}{s+2}+\frac{1}{s+3}$$

$$U_{Czi}(s)=\frac{2}{s^2+3s+2}\left(\frac{s-1}{s}\right)+\frac{1}{s}=\frac{2(s-1)}{s(s+1)(s+2)}+\frac{1}{s}$$

$$=\frac{-1}{s}+\frac{4}{s+1}-\frac{3}{s+2}+\frac{1}{s}=\frac{4}{s+1}-\frac{3}{s+2}$$

(3) Find the time-domain expression of the response by inverse Laplace transform.

$$u_{Czs}(t)=(e^{-t}-2e^{-2t}+e^{-3t})u(t) \qquad u_{Czi}(t)=(4e^{-t}-3e^{-2t})u(t)$$

$$u_C(t)=u_{C_{zs}}(t)+u_{C_{zi}}(t)=(5e^{-t}-5e^{-2t}+e^{-3t})u(t)$$

Homework

- Repeat the examples in the lecture notes.
- 6.37(a)(c), and 6.39(a) in the textbook.

Review of Last Lecture

4.5 Inverse Laplace Transform

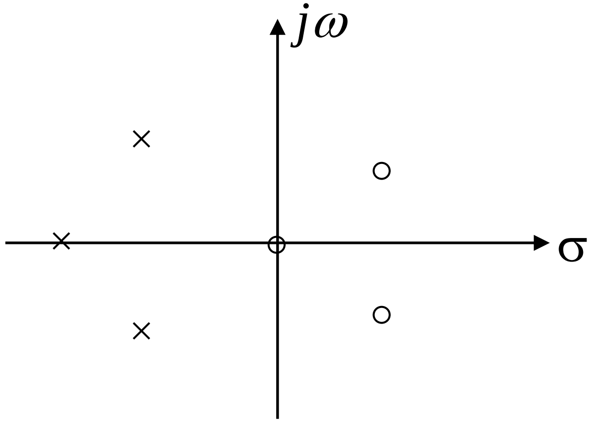
4.6 Analyzing Circuits and S-domain Models via Inverse Laplace Method

4.5.1 Zeros and Poles of Laplace Transform

$$f(t) = \mathcal{L}^{-1}\{F(s)\} = \frac{1}{2\pi j}\int_{\sigma - j\infty}^{\sigma + j\infty} F(s)e^{st}ds$$

According to the formula of the inverse Laplace transform, contour integration can be applied to solve it. However, in practice, most Laplace transforms are in the form of **rational fractions**:

$$F(s) = \frac{N(s)}{D(s)} = \frac{\sum_{i=0}^{m} b_i s^i}{\sum_{i=0}^{n} a_i s^i} = \frac{\prod_{i=1}^{m}(s - z_i)}{\prod_{i=1}^{n}(s - p_i)}$$



Zero-pole Distribution Diagram

The roots $\mathbf{z}_i$ of the numerator polynomial $N(s) = 0$ are called the **zeros** of ***F(s)***;

the roots $\boldsymbol{p}_i$ of the denominator polynomial $D(s) = 0$ are called the **poles** of ***F(s)***

4.5.2 Finding the Inverse Laplace Transform

When analyzing systems using the Laplace transform method, the final step is to perform the **inverse Laplace transform** on the image function.

1. Partial Fraction Method (Heaviside Method)

(1) Case of simple real poles $(p_1 \cdots p_n)$

When $m < n$, $F(s) \overset{Decomposition}{==} \dfrac{k_1}{s-p_1} + \cdots + \dfrac{k_n}{s-p_n}$ **(**Proper Fraction**)**

where $k_i = (s-p_i)F(s)\big|_{s=p_i}$ (Residue) (See the textbook for the proof)

It is known from the properties of Laplace transform：

$$Ae^{-\alpha t}u(t) \overset{LT}{\longleftrightarrow} \frac{A}{s+\alpha}$$

$$\boxed{F(s) \xrightarrow{L^{-1}} f(t) = \left(k_1 e^{p_1 t} + \cdots + k_n e^{p_n t}\right)u(t)}$$

(2) The case where poles include conjugate complex roots $(p_{1,2} = -\alpha \pm j\beta)$

$$F(s) = \frac{A(s)}{D(s)\left[(s+\alpha)^2 + \beta^2\right]}$$

$$= \frac{A(s)}{D(s)(s+\alpha-j\beta)(s+\alpha+j\beta)}$$

where $D(s)$ is the remaining part of the denominator after removing the conjugate complex roots.

Set $F_1(s) = \dfrac{A(s)}{D(s)}$

then $F(s) = \dfrac{F_1(s)}{(s+\alpha-j\beta)(s+\alpha+j\beta)}$

$$\overset{Decomposition}{=\!=} \frac{K_1}{s+\alpha-j\beta} + \frac{K_2}{s+\alpha+j\beta} + \dots$$

where $K_{1,2}=\dfrac{F_1(-\alpha\pm j\beta)}{\pm 2j\beta}$ (residue)

K_1、K_2 have a conjugate relationship: $K_{1,2}=A\pm jB$

Thus, the inverse transform of the part related to the conjugate complex roots

is $\xrightarrow{\mathrm{LT}^{-1}}$

$$
\begin{aligned}
f_c(t) &= e^{-\alpha t}(K_1 e^{j\beta t}+K_1^{*}e^{-j\beta t})u(t) \\
&= e^{-\alpha t}\{(A+jB)[\cos(\beta t)+j\sin(\beta t)]+(A-jB)[\cos(\beta t)-j\sin(\beta t)]\}u(t) \\
&= 2e^{-\alpha t}\left[A\cos(\beta t)-B\sin(\beta t)\right]u(t)
\end{aligned}
$$

α--attenuation factor; β--oscillation frequency

(3) The case where poles include multiple roots(a k-fold p_1, simple roots $p_2 \dots p_n$)

$$F(s)=\frac{A(s)}{D(s)}=\frac{A(s)}{(s-p_1)^k D_1(s)}$$

$$=\frac{K_{11}}{(s-p_1)^k}+\frac{K_{12}}{(s-p_1)^{k-1}}+\cdots+\frac{K_{1k}}{(s-p_1)}+\sum_{i=2}^{n}\frac{K_i}{s-p_i}$$

$$=\sum_{i=1}^{k}\frac{K_{1i}}{(s-p_1)^{k-i+1}}+\sum_{i=2}^{n}\frac{K_i}{s-p_i}$$

To find K_i and K_{11}, the method is the same as the first case:

$$K_{11}=(s-p_1)^k F(s)\Big|_{s=p_1}$$

To find other coefficients,

Let $F_1(s)=(s-p_1)^k\cdot F(s)$

$$\text{when } i=2,\quad K_{12}=\frac{\mathrm{d}}{\mathrm{d}s}F_1(s)\Big|_{s=p_1}$$

$$\text{when } i=3,\quad K_{13}=\frac{1}{2}\frac{\mathrm{d}^2}{\mathrm{d}s^2}F_1(s)\Big|_{s=p_1}$$

$$K_{1i} = \frac{1}{(i-1)!} \frac{\mathrm{d}^{i-1}}{\mathrm{d}\, s^{i-1}} F_1(s) \Big|_{s=p_1} \qquad (i = 1, 2, \cdots k)$$

$$F_1(s) = (s - p_1)^k \cdot F(s)$$

The inverse transform of the multiple root part $\xrightarrow{\mathrm{LT}^{-1}}$:

$$f_c(t) = e^{p_1 t}\left[\frac{K_{11}}{(k-1)!}t^{k-1} + \frac{K_{12}}{(k-2)!}t^{k-2} \cdots + \frac{K_{1i}}{(k-i)!}t^{k-i} \cdots + K_{1k}\right]u(t)$$

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Given $F(s)=\left(\frac{1-\mathrm{e}^{-s}}{s}\right)^2$, find its Inverse Laplace Transform $f(\mathrm{t})$.

A $f(t)=tu(t)-2(t-1)u(t-1)+(t-2)u(t-2)$

B $f(t)=(t-1)u(t-1)+(t-2)u(t-2)$

C $f(t)=tu(t)-(t-1)u(t-1)+(t-2)u(t-2)$

D $f(t)=tu(t)-(t-2)u(t-2)$

提交

Given $F(s)=\left(\frac{1-\mathrm{e}^{-s}}{s}\right)^2$, find its Inverse Laplace Transform $f\,(\mathrm{t})$.

Solution: $F(s)=\frac{1-2\mathrm{e}^{-s}+\mathrm{e}^{-2s}}{s^2}=\frac{1}{s^2}-\frac{2}{s^2}\mathrm{e}^{-s}+\frac{1}{s^2}\mathrm{e}^{-2s}$

$$tu(t)\leftrightarrow\frac{1}{s^2}$$

Using the Time-Shifting Property:

$$\therefore\ \ f(t)=tu(t)-2(t-1)u(t-1)+(t-2)u(t-2)$$

Even if ***F*(*s*)** is not a proper fraction, i.e. ***n*<*m*,** it can also be expressed as the sum of a polynomial in ***s*** and a **proper fraction**:

Example: $$F(s)=\frac{s^3+4s^2+6s+5}{s^2+3s+2}=s+1+\frac{s+3}{s^2+3s+2}$$

In the above equation, the inverse Laplace transform of ***s*+1** is equal to : $\delta'(t)+\delta(t)$

For the inverse transform of a **rational proper fraction** in Laplace transform, solve it using the **partial fraction method**.

If ***F*(*s*)** is an **irrational function** containing $e^{-\alpha s}$, the term $e^{-\alpha s}$ does not participate in partial fraction decomposition, and the solution utilizes the **Time-Shifting Property.**

Example: $$\frac{\mathrm{e}^{-2s}}{s^2+3s+2}=F_1(s)\mathrm{e}^{-2s} \qquad F_1(s)=\frac{1}{s+1}+\frac{-1}{s+2}$$

So $f_1(t)=L^{-1}\left[F_1(s)\right]=\left(\mathrm{e}^{-t}-\mathrm{e}^{-2t}\right)u(t)$

So $f(t)=f_1(t-2)=\left[\mathrm{e}^{-(t-2)}-\mathrm{e}^{-2(t-2)}\right]u(t-2)$

4.6.1 Solving Differential Equations Using Laplace Transform

Steps for Solving Differential Equations Using Laplace Transform:

(1) Take the Laplace transform of the differential equation, paying attention to applying the differentiation property;

(2) Solve the algebraic equation in the s-domain to find the Laplace transform of the response;

(3) Find the inverse Laplace transform, i.e., find the time-domain expression of the response.

4.6.2 Solving Circuits Using Laplace Transform

s-domain Circuit Analysis Method

1. Replace each component in the network with its **s-domain model;**
Write the signal source as a transformed expression ($\frac{E}{s}$ or $\frac{E}{sR}$, $U_s(s)$ or $I_s(s)$)

2. Analyze the constructed s-domain model diagram using KVL and KCL to obtain the required transformed system equations.
(The numerical operations performed are algebraic relationships, similar to resistive networks; Thevenin's theorem and Norton's theorem are both applicable; It is suitable for network analysis with many nodes or loops.)

3. Find the **Laplace transform** of the response.

4. Find the **inverse Laplace transform**, i.e., find the time-domain expression of the response.

4.6.2 Solving Circuits Using Laplace Transform

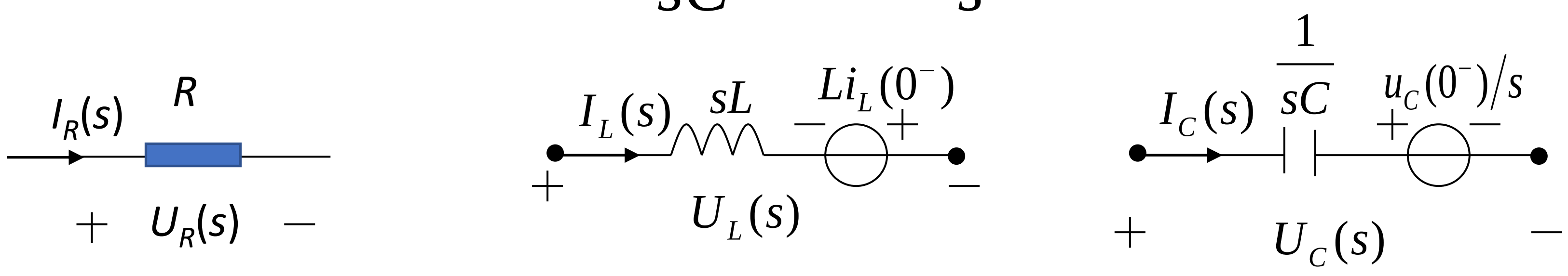
$s-$domain Component Models:

1) Loop Analysis (KVL)

$$U_R(s) = RI_R(s)$$

$$U_L(s) = sLI_L(s) - Li_L(0^-)$$

$$U_C(s) = \frac{1}{sC} I_C(s) + \frac{1}{s} u_c(0^-)$$

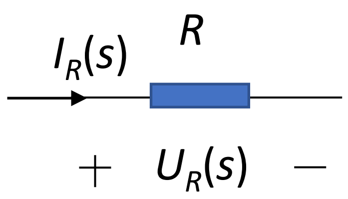


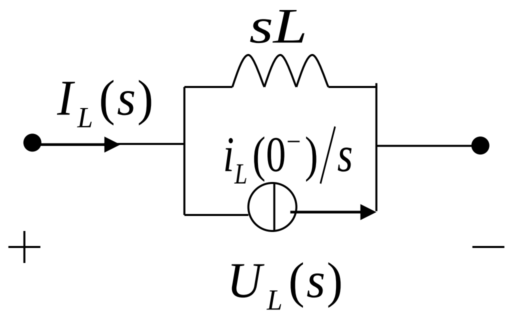
2) Node Analysis (KCL)

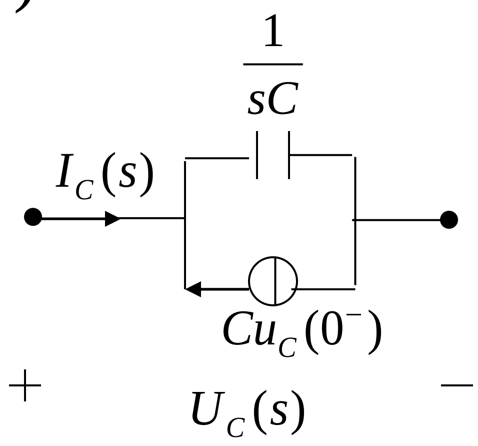
$$I_R = \frac{1}{R} U_R(s)$$

$$I_L(s) = \frac{1}{sL} U_L(s) + \frac{1}{s} i_L(0^-)$$

$$I_C(s) = sCU_C(s) - Cu_c(0^-)$$







Contents of This Lecture

Learning Objectives of This Lecture

1. Proficiently grasp the interrelationship among **system function**, **system differential equation**, and **system simulation block diagram** (**Important**);
2. Be able to qualitatively determine the characteristics of **impulse response** using the zero-pole distribution of **system function**;
3. Proficiently use **system function** to analyze the time-domain characteristics of **zero-state response**;
4. Proficiently use the zero-pole distribution of **system function** to roughly sketch the **frequency response characteristic** curve (**Important**).

4.7 System Function

4.7.1 Definition of System Function

If a time signal acts on a system and its output is still this time signal, only the **amplitude and phase are changed**, this time signal is called the **characteristic signal** of the system. The function that characterizes the changed amplitude and phase is called the **eigenvalue** or **system function** of the system.

The signal $e^{j\omega t}$ is the **characteristic signal** of the system, and the function $H(\omega)$ (the Fourier transform of the system's unit impulse response ***h*(*t*)** is the **system function**, also known as the **frequency response** of the system.

$$e^{j\omega t} \longrightarrow \boxed{h(t)} \longrightarrow \quad y(t) = h(t) * e^{j\omega t} = H(\omega)e^{j\omega t} = \left|H(\omega)\right| e^{j[\omega t + \varphi(\omega)]}$$

In fact, the **exponential signal** e^{st} is also a **characteristic signal** of the system.

$$e^{st} \longrightarrow \boxed{h(t)} \longrightarrow \quad y(t) = h(t) * e^{st} = H(s)e^{st}$$

$$y(t) = h(t) * e^{st} = \int_{0_-}^{\infty} h(\tau)e^{s(t-\tau)}d\tau = e^{st}\int_{0_-}^{\infty} h(\tau)e^{-s\tau}d\tau = H(s)e^{st}$$

4.7 System Function

$H(s)=\int_{0^-}^{\infty} h(t)e^{-st}dt$ is the Laplace transform of the unit impulse response of a **causal system**, also known as the **system function**.

For a continuous-time signal ***x(t)***, acting on a linear time-invariant system with unit impulse response ***h(t)***, its **zero-state response** $y_{zs}(t)$ is equal to the **convolution** of the input signal and the unit impulse response:

$$y_{zs}(t)=h(t)*x(t)$$

$$x(t) \longrightarrow \boxed{h(t)} \longrightarrow y_{zs}(t)=h(t)*x(t)$$

By the **time-domain convolution theorem**, the Laplace transform of the zero-state response is equal to the **product** of the Laplace transform of the input signal and the Laplace transform of the unit impulse response:

$$Y_{zs}(s)=H(s)X(s)$$

$$X(s) \longrightarrow \boxed{H(s)} \longrightarrow Y_{zs}(s)=H(s)X(s)$$

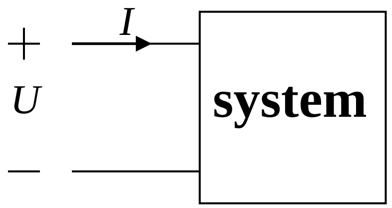
The **system function** is equal to the **ratio** of the Laplace transform of the zero-state response to the Laplace transform of the input signal:

$$\boxed{H(s)=Y_{zs}(s)/X(s)}$$

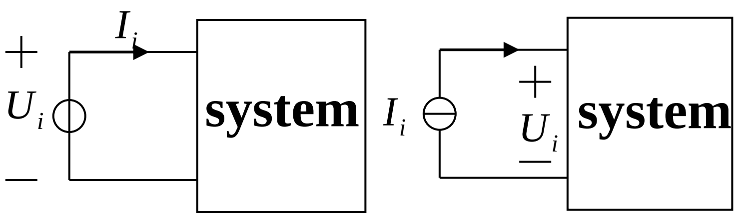
4.7.2 Driving Point Function and Transfer Function

1. Driving Point Function

When the input and output of the system are taken from the same port, the system function is called the **driving point function**.

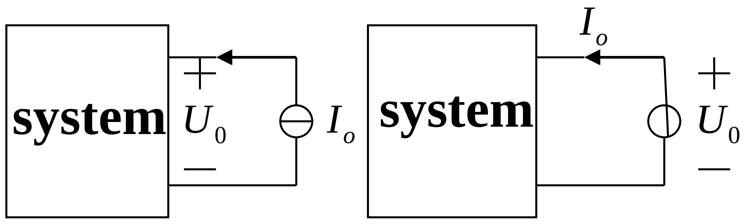


Depending on its dimension, the driving point function can also be called **impedance function** or **admittance function**. For example, the input or output impedance (ratio of voltage to current) of an amplifier, input or output admittance (ratio of current to voltage), etc.



Impedance: $Z_i(s) = \dfrac{U_i(s)}{I_i(s)}$ $\quad Z_o(s) = \dfrac{U_o(s)}{I_o(s)}$

Admittance: $Y_i(s) = \dfrac{I_i(s)}{U_i(s)}$ $\quad Y_o(s) = \dfrac{I_o(s)}{U_o(s)}$

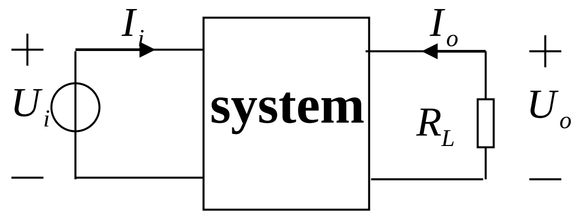


2. Transfer (Transmission) Function

When the input and output of the system are taken from different ports, the system function is called the **transfer function** or **transmission function**.

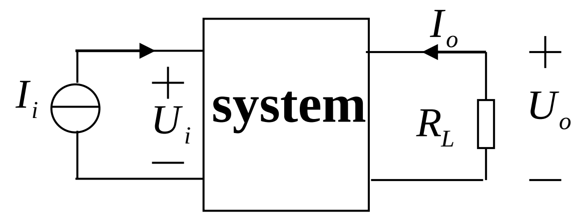
Depending on its dimension, the transfer function can be called **transfer impedance**, **transfer admittance**, **voltage transmission ratio**, or **current transmission ratio**, etc. For example, the voltage or current gain of an amplifier.

Transfer Impedance: $Z_{21}(s)=\dfrac{U_o(s)}{I_i(s)}$



Transfer Admittance: $Y_{21}(s)=\dfrac{I_o(s)}{U_i(s)}$

Voltage Transmission Ratio: $K_v(s)=\dfrac{U_o(s)}{U_i(s)}$



Current Transmission Ratio: $K_i(s)=\dfrac{I_o(s)}{I_i(s)}$

4.7.3 Solution of System Function

Since the system function is the ratio of the Laplace transform of the system's zero-state response to that of the excitation, **its solution is independent of the system's initial conditions**.

Example 4-23: Given the system's differential equation :

$$\frac{d^2y(t)}{dt^2}+3\frac{dy(t)}{dt}+2y(t)=\frac{dx(t)}{dt}+3x(t)$$

find the system function $\boldsymbol{H(s)}$ and the system's unit impulse response $\boldsymbol{h(t)}$.

Solution: Take the Laplace transform of the above equation. When using the differential property, **the initial conditions are zero.**

$$(s^2+3s+2)Y(s)=(s+3)X(s)$$

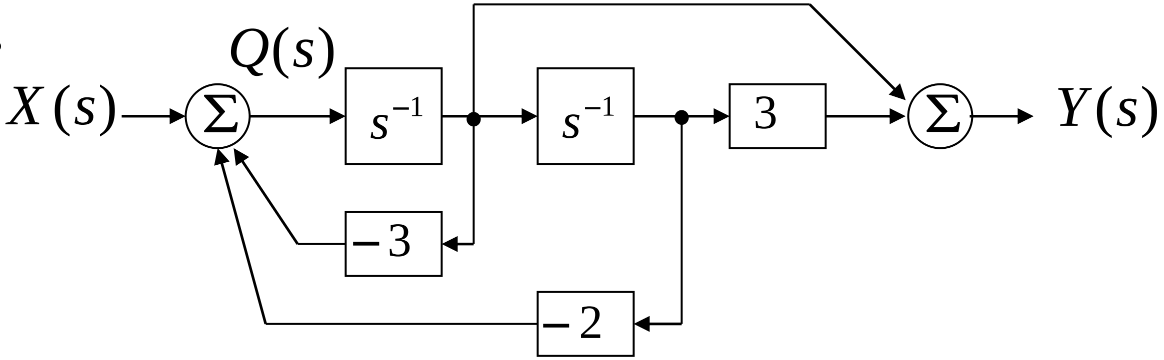
Thus, the system function is

$$H(s)=\frac{Y(s)}{X(s)}=\frac{s+3}{s^2+3s+2}=\frac{s+3}{(s+1)(s+2)}=\frac{2}{s+1}-\frac{1}{s+2}$$

The unit impulse response of the system is obtained by taking the inverse Laplace transform of the system function:

$$h(t)=(2e^{-t}-e^{-2t})u(t)$$

Example 4-24: Given the simulation block diagram of a system as follows, find the system function ***H*(*s*)**.



Solution: Let ***Q*(*s*)** be the output of the previous adder. Then:

$$Q(s) = X(s) - 3s^{-1}Q(s) - 2s^{-2}Q(s)$$

$$X(s) = (1 + 3s^{-1} + 2s^{-2})Q(s)$$

The output ***Y*(*s*)** of the latter adder is equal to:

$$Y(s) = s^{-1}Q(s) + 3s^{-2}Q(s) = (s^{-1} + 3s^{-2})Q(s)$$

thus,

$$H(s) = \frac{Y(s)}{X(s)} = \frac{s^{-1} + 3s^{-2}}{1 + 3s^{-1} + 2s^{-2}} = \frac{s+3}{s^2 + 3s + 2}$$

4.8.1 Zero-pole Distribution of System Function and Waveform of Unit Impulse Response

The unit impulse response of a system is the inverse Laplace transform of the system function: $h(t) = L^{-1}\{H(s)\}$

Generally, the system function is a rational fraction, so:

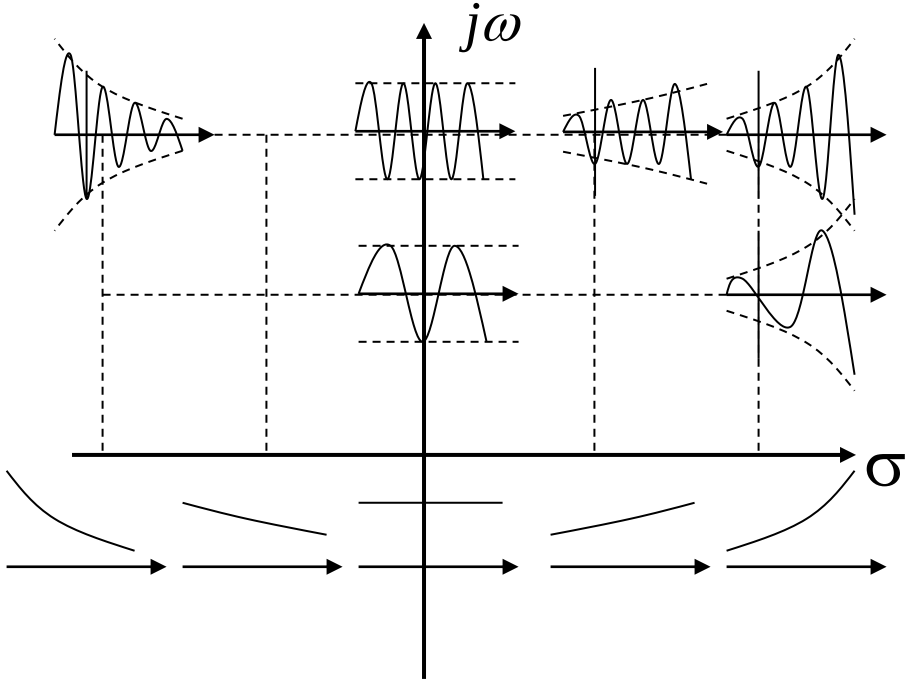
$$h(t) = L^{-1}\{\frac{N(s)}{D(s)}\} = L^{-1}\{\frac{A\prod_{k=1}^{M}(s-z_k)}{\prod_{k=1}^{N}(s-p_k)}\} = L^{-1}\{\sum_{k=1}^{N}\frac{A_k}{s-p_l}\} = \sum_{k=1}^{N} A_k e^{p_k t}u(t)$$

In the above equation, $\boldsymbol{z_k}$ and $\boldsymbol{p_k}$ are the zeros and poles of the system function, respectively. The different positions of zeros and poles in the s-plane have different effects on the unit impulse response.

The **s**-plane is a complex plane, where the ***σ***-axis is the real axis and the imaginary axis is denoted by ***jω***. Different points ***s*** on the plane correspond to different time signals $\boldsymbol{e^{st}}$.

$$e^{st}=e^{(\sigma+j\omega)t}=e^{\sigma t}\cdot e^{j\omega t}=e^{\sigma t}(\cos\omega t+j\sin\omega t)$$

The signal e^{st} represents a sine-cosine signal with amplitude changing exponentially. **The farther a point is from the real axis, the higher the frequency of the signal change; the farther a point is from the imaginary axis, the faster the signal amplitude grows or decays.**



Points on the real axis represent signals with monotonic exponential changes;

points on the imaginary axis represent sine-cosine signals with constant-amplitude oscillations.

First-order Poles

$$H(s)=\frac{1}{s},\qquad p_1=0 \text{ at the origin},\quad h(t)=L^{-1}[H(s)]=u(t)$$

$$H(s)=\frac{1}{s+\alpha},\qquad p_1=-\alpha$$

$\alpha>0$, on the left real axis, $h(t)=\mathrm{e}^{-\alpha t}u(t)$, exponential decay

$\alpha<0$, on the right real axis, $h(t)=\mathrm{e}^{-\alpha t}u(t)$, exponential growth

$$H(s)=\frac{\omega}{s^2+\omega^2},\qquad p_{1,2}=\pm \mathrm{j}\omega,\ \text{on the imaginary axis}$$

$h(t)=\sin(\omega t)u(t)$, constant – amplitude oscillation

$$H(s)=\frac{\omega}{(s+\alpha)^2+\omega^2},\qquad p_{1,2}=-\alpha\pm \mathrm{j}\omega \qquad h(t)=e^{-\alpha t}\sin\omega t u(t)$$

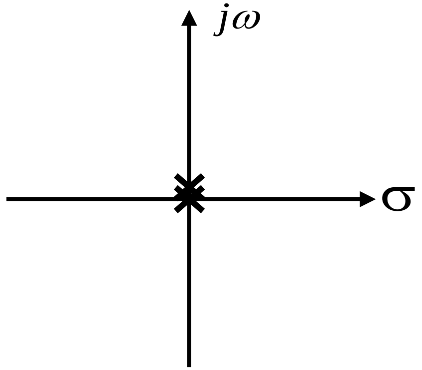
When $\alpha>0$, poles in the left half-plane, damped oscillation;
When $\alpha<0$, poles in the right half-plane, growing oscillation

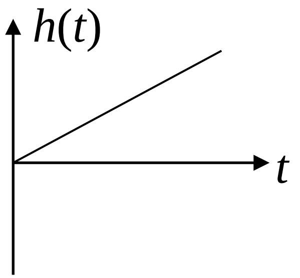
Second-order Poles

$$H(s)=\frac{1}{s^2}\text{, pole at the origin,}$$

$$h(t)=tu(t)$$

$$t\to\infty, h(t)\to\infty$$

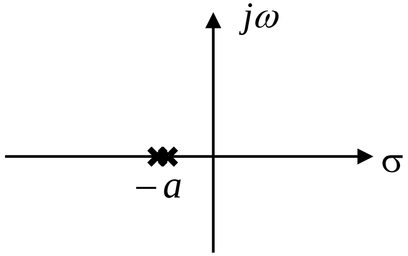


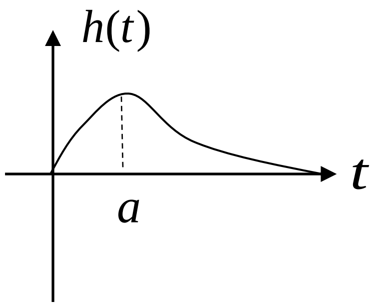


$$H(s)=\frac{1}{(s+\alpha)^2}\text{, pole on the real axis,}$$

$$h(t)=t\,\mathrm{e}^{-\alpha t}u(t), \alpha>0$$

$$t\to\infty, h(t)\to 0$$

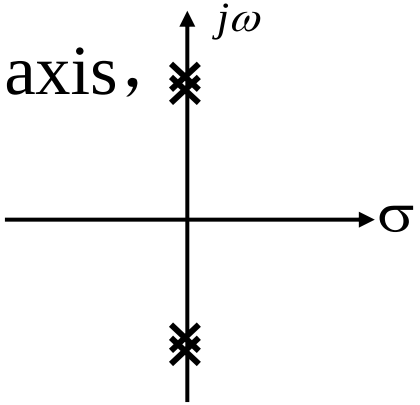


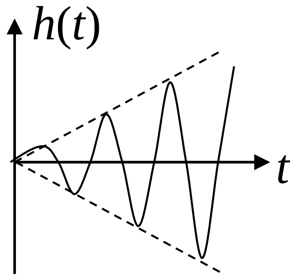


$$H(s)=\frac{2\omega s}{(s^2+\omega^2)^2}\text{, pole on the imaginary axis,}$$

$$h(t)=t\sin\omega t u(t)$$

$t\to\infty, h(t)$ exhibits growing oscillation

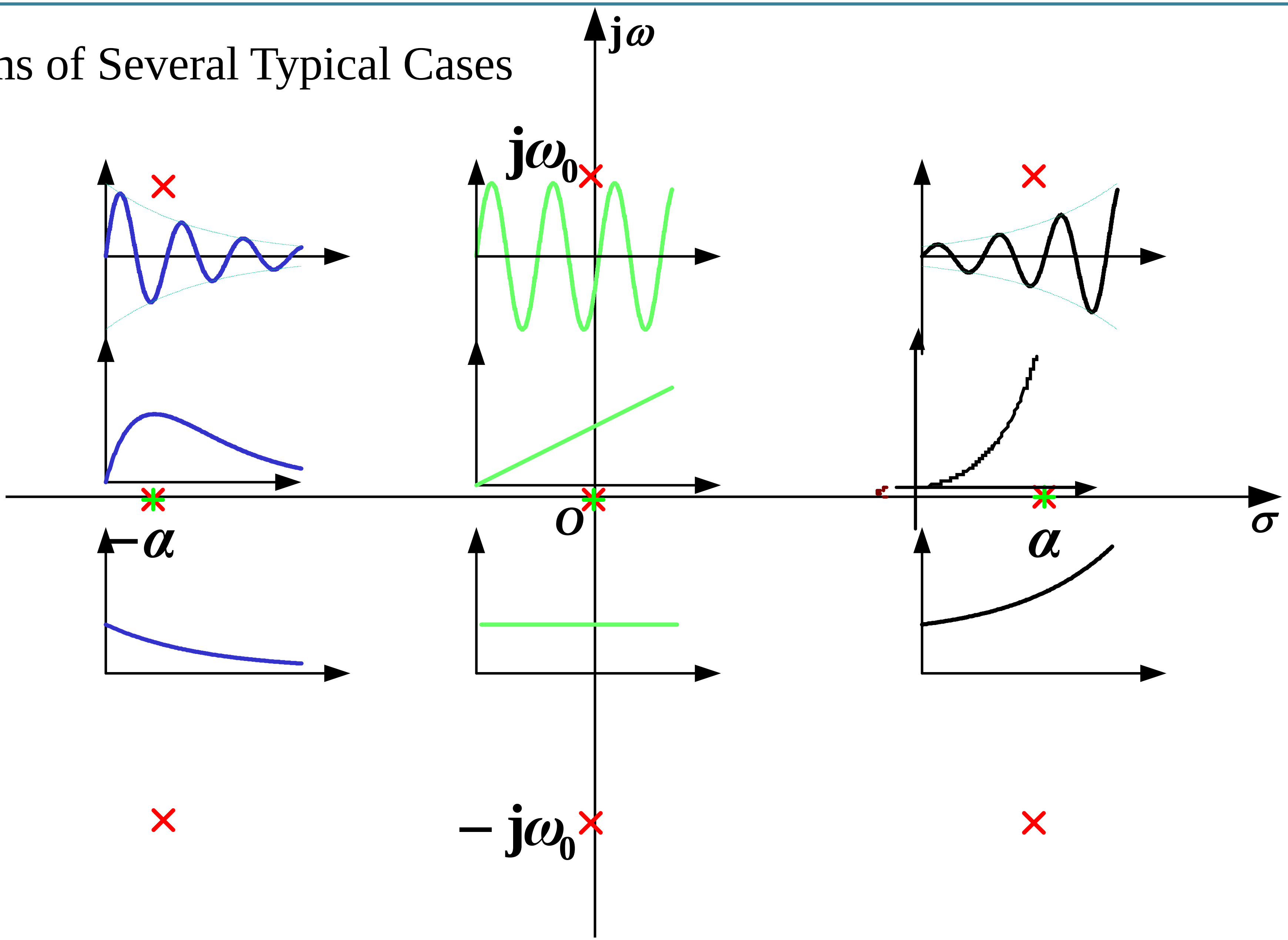




Correspondence between Poles of $\boldsymbol{H(s)}$ and Waveform Characteristics of $\boldsymbol{h(t)}$:

Poles on the left half-plane of the s-plane	**$h(t)$ exhibits a decaying form**
Poles on the right half-plane of the s-plane	$\boldsymbol{h(t)}$ exhibits a growing form
Poles on the imaginary axis of the s-plane (excluding the origin)	$\boldsymbol{h(t)}$ has constant-amplitude oscillation (first-order poles) or growing oscillation (multi-order poles)
Poles on the real axis of the s-plane (excluding the origin)	$\boldsymbol{h(t)}$ exhibits exponential-related changes (first-order or multi-order poles)
Poles at the origin of the s-plane	$\boldsymbol{h(t)}$ is equal to $\boldsymbol{u(t)}$(first-order pole) or grows as t^n (multi-order poles)

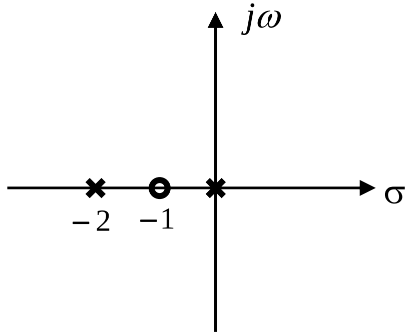
Diagrams of Several Typical Cases



Example 4-25: Given the following system functions, try to draw their zero-pole plots, find their unit impulse responses, and sketch their waveforms.

$$H_1(s)=\frac{s+1}{s(s+2)} \qquad H_2(s)=\frac{s+1}{s^2+2s+2} \qquad H_3(s)=\frac{1}{(s+1)^2}$$

Solution: The zero-pole plot of $\boldsymbol{H_1(s)}$ is as follow：

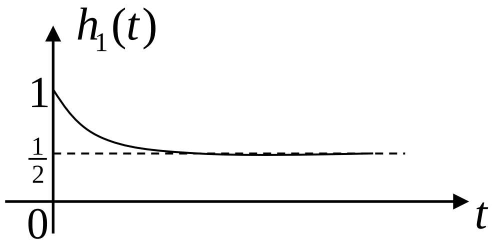


Because,

$$H_1(s)=\frac{s+1}{s(s+2)}=\frac{1}{2s}+\frac{1}{2(s+2)}$$

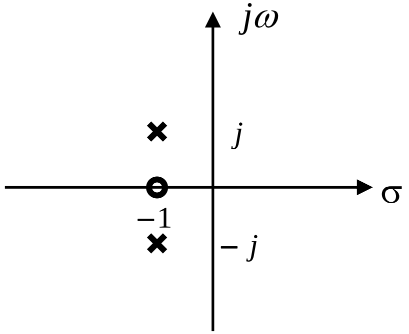
Thus,

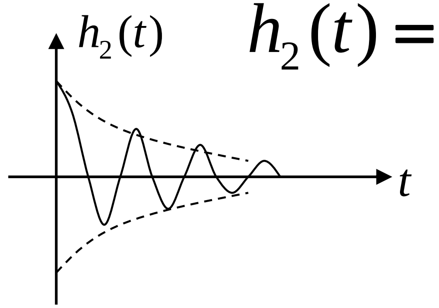
$$h_1(t)=\frac{1}{2}(1+e^{-2t})u(t)$$



The zero-pole plot of $\boldsymbol{H_2(s)}$ is as follow：

$$H_2(s)=\frac{s+1}{s^2+2s+2}=\frac{s+1}{(s+1)^2+1}$$

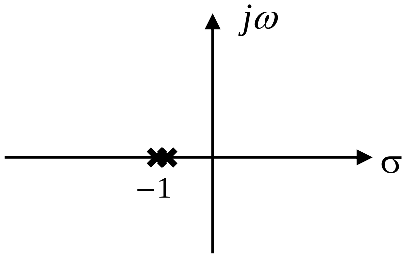


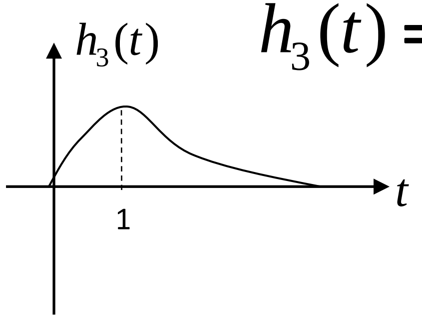


$$h_2(t)=e^{-t}\cos t u(t)$$

The zero-pole plot of $\boldsymbol{H_3(s)}$ is as follow：

$$H_3(s)=\frac{1}{(s+1)^2}$$





$$h_3(t)=te^{-t}u(t)$$

It can be seen from the above example that if a pole is a multi-order pole, its corresponding time waveform is different from that of a first-order pole.

4.8.2 Zero-pole Distribution of System Function & Excitation and Response Components

The **poles** of the system function and the excitation jointly determine the waveform of the response. $Y(s) = X(s)H(s)$

Example 4-26: Let the system function be $H(s) = \dfrac{s+3}{s^2+3s+2}$, the system excitation be $x(t) = u(t)$, and the initial state be zero. Find the zero-state response of the system, and identify the free response and forced response.

Solution: The Laplace transform of the zero-state response of the system is

$$Y(s) = X(s)H(s) = \frac{1}{s}\cdot\frac{s+3}{s^2+3s+2} = \frac{s+3}{s(s+1)(s+2)} = \frac{\frac{3}{2}}{s} - \frac{2}{s+1} + \frac{\frac{1}{2}}{s+2}$$

Therefore,

the zero-state response of the system is $y(t) = \dfrac{1}{2}(3 - 4e^{-t} + e^{-2t})u(t)$

where the free response is $y_h(t) = (\dfrac{1}{2}e^{-2t} - 2e^{-t})u(t)$, and the forced response is $y_p(t) = \dfrac{3}{2}u(t)$

The poles of the system function correspond to the free response components; the poles of the excitation correspond to the forced response components.

Example 4-27： Given the differential equation of an LTI system:

$$r''(t)+2r'(t)+r(t)=e'(t)$$

Find the zero-state response $r_{zs}(t)$ if the excitation is $e(t)=u(t)$.

Solution： Take the Laplace transform of both sides of the differential equation:

$$(s^2+2s+1)R(s)=sE(s)$$

$$H(s)=\frac{R(s)}{E(s)}=\frac{s}{s^2+2s+1}=\frac{s^{-1}}{1+2s^{-1}+s^{-2}}$$

Find the Laplace transform of the excitation: $E(s)=\frac{1}{s}$

zero-pole cancellation

$$R_{zs}(s)=H(s)E(s)=\frac{sE(s)}{s^2+2s+1}=\frac{s}{s(s^2+2s+1)}=\frac{1}{s^2+2s+1}=\frac{1}{(s+1)^2}$$

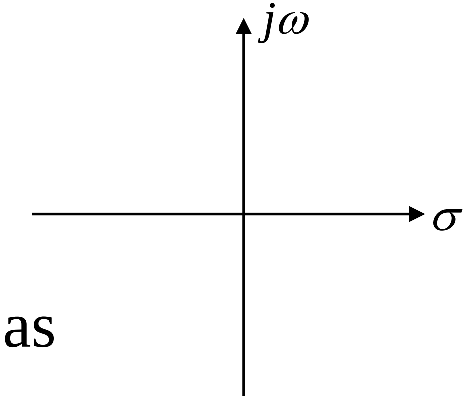
Using the Laplace transform of $tu(t)$ (which is $1/s^2$) and the frequency-shifting property, find the inverse Laplace transform of $R_{zs}(s)$:

$$r_{zs}(t)=te^{-t}u(t)$$

4.9 Frequency Response from System Function Zero-pole Distribution

4.9.1 $H(s)$ and $H(\omega)$

The Laplace transform is an extension of the Fourier transform from the real frequency domain **ω** to the complex frequency domain **s**， and the Fourier transform is a special case of the Laplace transform on the imaginary axis of the **s**-plane. **Generally speaking,** $H(\omega)=H(s)|_{s=j\omega}$



4.9.2 Zero-pole Distribution of $H(s)$ and $H(\omega)$

Since $\boldsymbol{H(s)}$ is generally a rational fraction, it can be expressed as

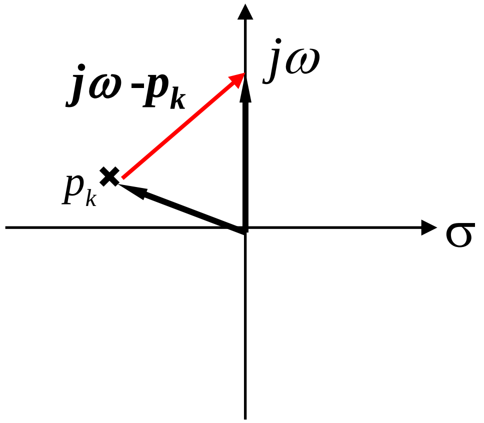
$$H(s)=\frac{N(s)}{D(s)}=\frac{G\prod_{k=1}^{M}(s-z_k)}{\prod_{k=1}^{N}(s-p_k)}$$

***G*—constant**

The frequency response of a system can **generally** be expressed as (for conditions, refer to Section 4.13 of the textbook):

$$H(\omega)=H(s)\big|_{s=j\omega}=\left.\frac{G\prod_{k=1}^{M}(s-z_k)}{\prod_{k=1}^{N}(s-p_k)}\right|_{s=j\omega}=\frac{G\prod_{k=1}^{M}(j\omega-z_k)}{\prod_{k=1}^{N}(j\omega-p_k)}$$

In the above equation, each factor in the numerator and denominator represents a vector in the **s-plane** pointing to the ***jω*-axis.**



As shown in the figure, the factor **($j\omega$ -p_k)** represents the difference vector between the vector ***jω*** (varying along the imaginary axis) and the vector from the origin to p_k.

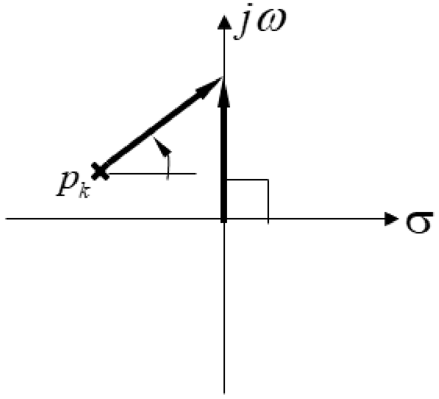
The vector corresponding to the factor in the numerator is called the **zero vector**;

the vector corresponding to the factor in the denominator is called the **pole vector**.

Each vector has its magnitude and phase, so:

$$H(\omega)=\frac{G\prod_{k=1}^{M}(j\omega-z_k)}{\prod_{k=1}^{N}(j\omega-p_k)}=\frac{G\prod_{k=1}^{M}B_k e^{j\beta_k}}{\prod_{k=1}^{N}A_k e^{j\theta_k}}=\frac{G\prod_{k=1}^{M}B_k}{\prod_{k=1}^{N}A_k}e^{j(\sum_{k=1}^{M}\beta_k-\sum_{k=1}^{N}\theta_k)}$$

$$=|H(\omega)|e^{j\varphi(\omega)}$$

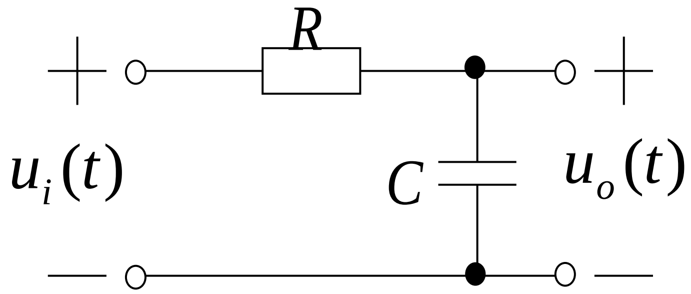


$$|H(\omega)|=\frac{G\prod_{k=1}^{M}B_k}{\prod_{k=1}^{N}A_k} \qquad \varphi(\omega)=\sum_{k=1}^{M}\beta_k-\sum_{k=1}^{N}\theta_k$$

The **magnitude-frequency response** of the system is **the ratio of the product of the magnitudes of zero vectors to the product of the magnitudes of pole vectors**.

The **phase-frequency response** of the system is **the sum of the phases of zero vectors minus the sum of the phases of pole vectors**.

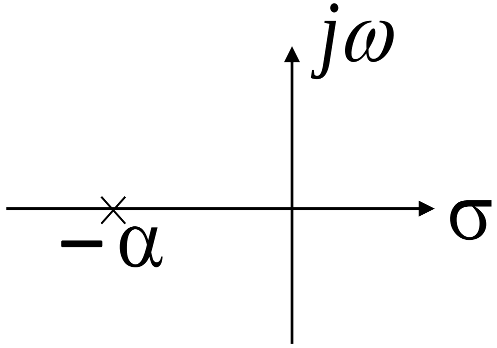
Example 4-28： For the RC circuit shown below, find its system function, draw its zero-pole plot, find its frequency response, and roughly sketch its frequency response curve.



Solution： The system function of the circuit is

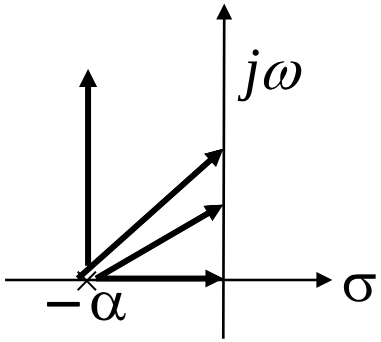
$$H(s)=\frac{U_o(s)}{U_i(s)}=\frac{\frac{1}{sC}}{R+\frac{1}{sC}}=\frac{1}{RCs+1}$$

$$=\frac{\frac{1}{RC}}{s+\frac{1}{RC}}=\frac{\alpha}{s+\alpha}\qquad\left(\alpha=\frac{1}{RC}\right)$$



4.9 Frequency Response from System Function Zero-pole Distribution

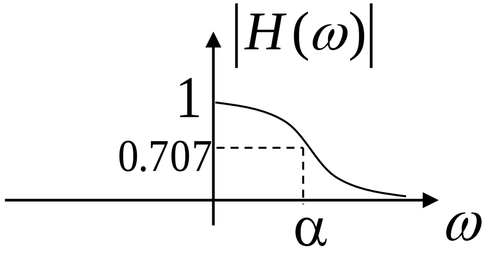
Roughly sketch the frequency response from $H(s)=\dfrac{\alpha}{s+\alpha}$:



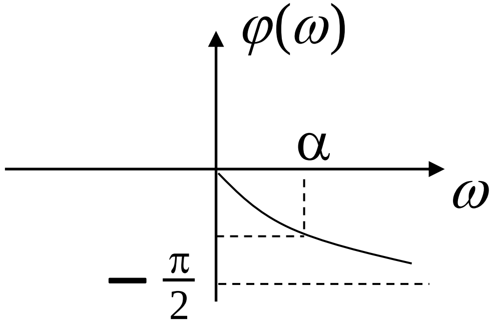
(1) when $\omega=0$, the pole vector points to the origin, with magnitude $\boldsymbol{\alpha}$ and phase 0; thus, $\left|H(\omega)\right| = \alpha/\alpha = 1$, $\varphi(\omega) = 0$

(2) when $\omega\uparrow$, the magnitude of the pole vector↑, phase↑; $\left|H(\omega)\right|\downarrow$, $\varphi(\omega)\downarrow$.

(3) when $\omega=\alpha$, $\left|H(\omega)\right| \approx 0.707$, $\varphi(\omega) = -\pi/4$.



(4) when $\omega\to\infty$, $\left|H(\omega)\right| \to 0$, $\varphi(\omega) \to -\pi/2$.



From its magnitude-frequency response curve, it can be seen that it is a **low-pass filter**.

$$H(\omega)=\frac{\alpha}{j\omega+\alpha} = \frac{\alpha}{\sqrt{\omega^2+\alpha^2}}\, e^{-j\,arctg(\frac{\omega}{\alpha})}$$

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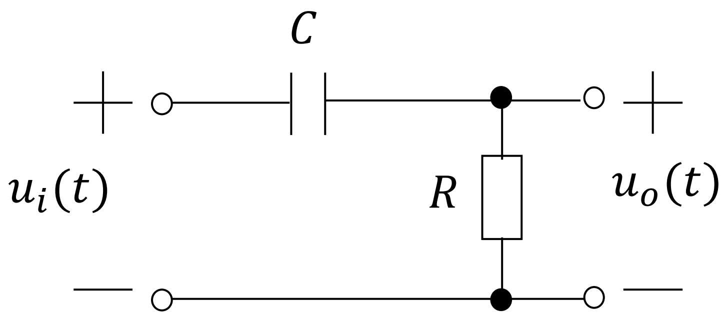
设置

For the RC circuit shown below, it represents a （ ）.



A Low-pass filter

B High-pass filter

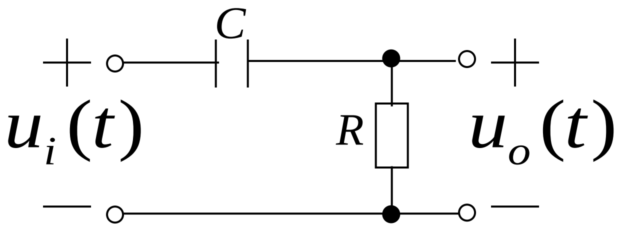
C Band-pass filter

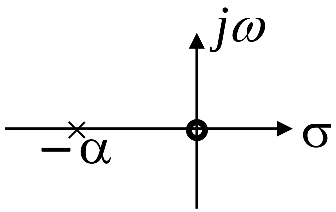
D All-pass filter

提交

4.9 Frequency Response from System Function Zero-pole Distribution

Example 4-29: For the RC circuit shown in the figure, find its system function, draw its zero-pole plot, find its frequency response, and roughly sketch its frequency response curve.





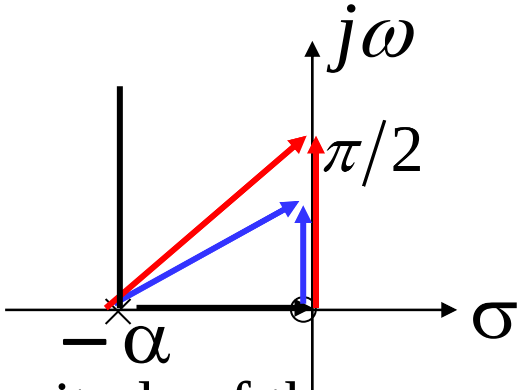
Solution: The system function of the circuit is derived as follows:

$$H(s)=\frac{U_o(s)}{U_i(s)}=\frac{R}{R+\dfrac{1}{sC}}=\frac{RCs}{RCs+1}$$

$$=\frac{s}{s+\dfrac{1}{RC}}=\frac{s}{s+\alpha}\qquad\left(\alpha=\frac{1}{RC}\right)$$

$$H(\omega)=\frac{j\omega}{j\omega+\alpha}$$

Roughly sketch the frequency response from $H(s)=\dfrac{s}{s+\alpha}$:



(1) when $\omega=0$, the pole vector points to the origin, with a magnitude of $\boldsymbol{\alpha}$ and a phase of **0;** the zero vector has a magnitude of 0 and a phase of $\boldsymbol{\pi}$**/2;** thus, $|H(\omega)|=0$, $\varphi(\omega)=\pi/2$.

(2) when $\boldsymbol{\omega}\uparrow$, the magnitude of the pole vector ↑, the phase↑; the magnitude of the zero vector ↑, **and its increasing rate is higher than that of the pole vector,** with a phase of $\boldsymbol{\pi}$**/2;** $|H(\omega)|\uparrow,\varphi(\omega)\downarrow=(\pi/2)-arctg(\omega/\alpha)$

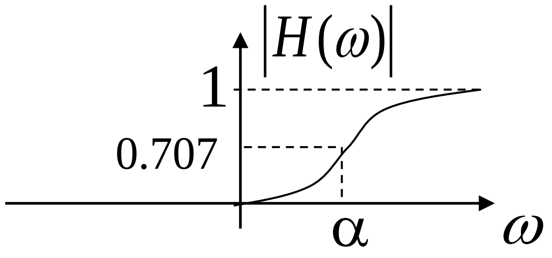
For example, when the phase is $\boldsymbol{\pi}$**/6,** the magnitude of the zero vector is $\sqrt{3}\alpha/3$, the magnitude of the pole vector is $2\sqrt{3}\alpha/3$, and $|H(\omega)|=1/2$; when the phase approaches **π/4,** the magnitude of the zero vector approaches $\boldsymbol{\alpha}$, the magnitude of the pole vector approaches $\sqrt{2}\alpha$, and $|H(\omega)|\to 0.707$.

(3) when $\omega=\alpha$, $|H(\omega)|\approx 0.707$, $\varphi(\omega)=\pi/4$.

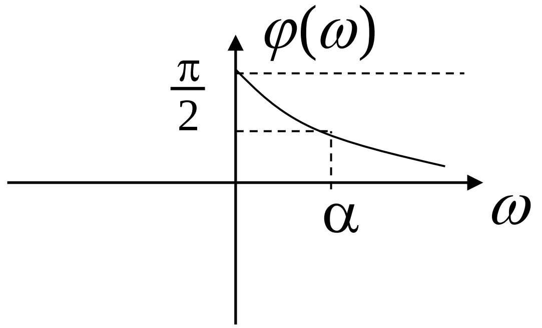
(4) when $\omega\to\infty$, both the zero vector and the pole vector are perpendicular to the real axis and point to infinity, with infinite magnitude and a phase of **π/2.**

At this point, $|H(\omega)|\to 1$, $\varphi(\omega)\to 0$.

$$|H(\omega)| = \frac{|\omega|}{\sqrt{\omega^2+\alpha^2}}$$



$$\varphi(\omega) = \begin{cases} \frac{\pi}{2} - arctg(\frac{\omega}{\alpha}) & \omega > 0 \\ -\frac{\pi}{2} - arctg(\frac{\omega}{\alpha}) & \omega < 0 \end{cases}$$



From its magnitude-frequency response curve, it can be seen that this is a **high-pass filter**.

$$H_{HP}(\omega) = \frac{j\omega}{j\omega+\alpha}, \quad H_{LP}(\omega) = \frac{\alpha}{j\omega+\alpha}$$

$$H_{HP}(\omega) = 1 - H_{LP}(\omega)$$

Homework

待定

Review of Last Lecture

4.7 System Function

4.8 Time-domain Characteristics from System Function Zero-pole Distribution

4.9 Frequency Response from System Function Zero-pole Distribution

4.7 System Function

$H(s)=\int_{0^-}^{\infty} h(t)e^{-st}dt$ is the Laplace transform of the unit impulse response of a **causal system**, also known as the **system function**.

For a continuous-time signal ***x*(*t*)**， acting on a linear time-invariant system with unit impulse response ***h*(*t*)**, its **zero-state response** $\boldsymbol{y_{zs}(t)}$ is equal to the **convolution** of the input signal and the unit impulse response:

$$x(t) \longrightarrow \boxed{h(t)} \longrightarrow y_{zs}(t)=h(t)*x(t)$$

By the **time-domain convolution theorem**, the Laplace transform of the zero-state response is equal to the **product** of the Laplace transform of the input signal and the Laplace transform of the unit impulse response:

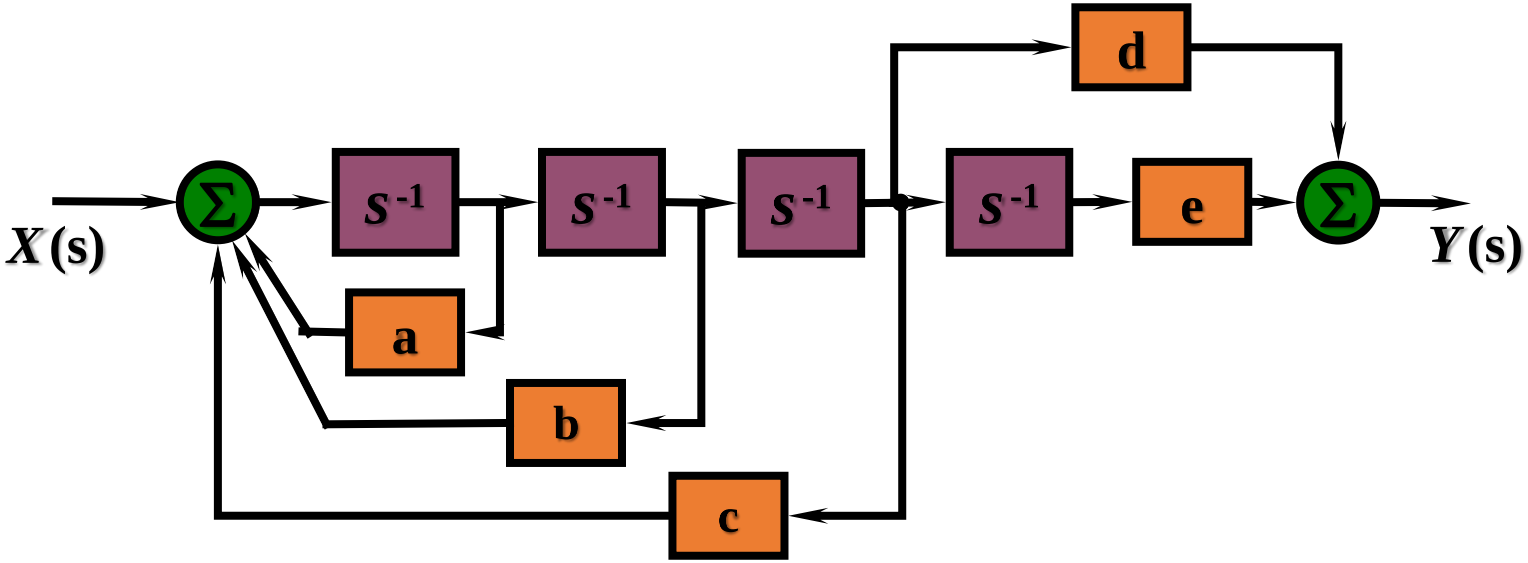
$$X(s) \longrightarrow \boxed{H(s)} \longrightarrow Y_{zs}(s)=H(s)X(s)$$

The **system function** is equal to the **ratio** of the Laplace transform of the zero-state response to the Laplace transform of the input signal:

$$\boxed{H(s)=Y_{zs}(s)/X(s)}$$

Given $H(s)=\dfrac{2s+3}{s(s+3)(s+2)^2}$, find the parameters in the s-domain block diagram:

$a=$ [填空1] ,$b=$ [填空2], $c=$ [填空3] , $d=$ [填空4] , $e=$ [填空5] .

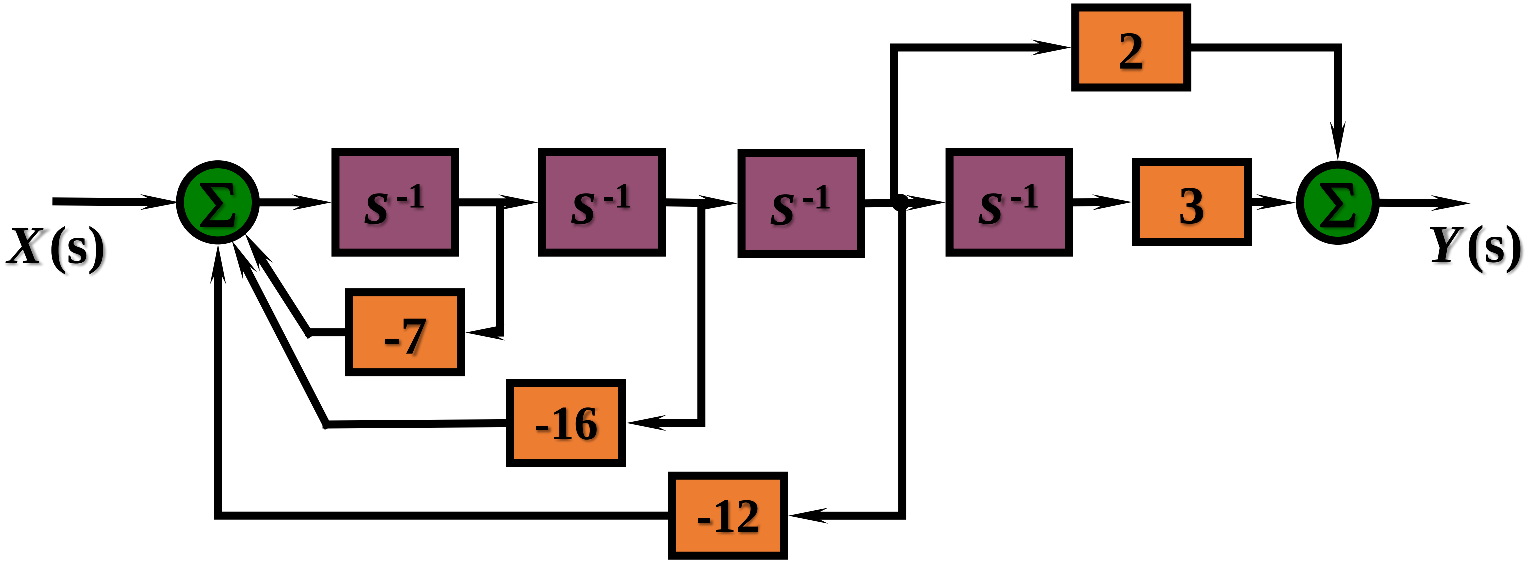


作答

Given $H(s)=\dfrac{2s+3}{s(s+3)(s+2)^2}$, find the parameters in the s-domain block diagram.

Solution: The system function can be transformed into:

$$H(s)=\frac{2s+3}{s(s+3)(s+2)^2}=\frac{2s+3}{s^4+7s^3+16s^2+12s}=\frac{2s^{-3}+3s^{-4}}{1+7s^{-1}+16s^{-2}+12s^{-3}}$$



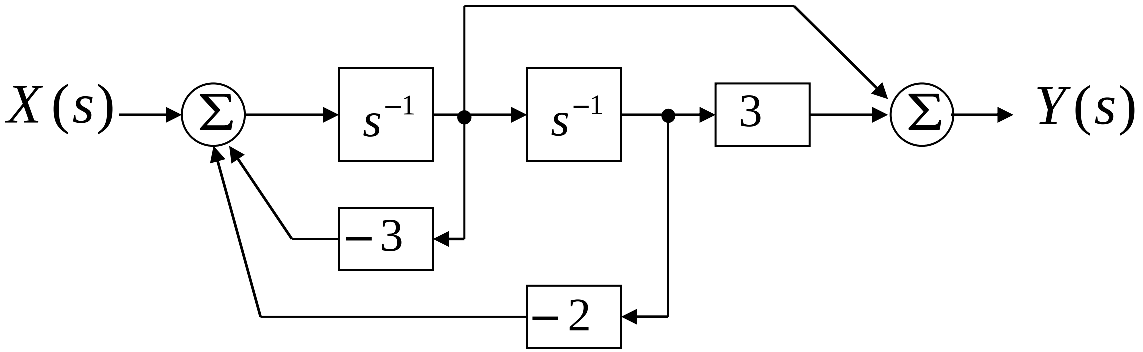
4.7 System Function

Roles of System Function

1. To find the **zero-state response** of the system： $Y_{zs}(s) = H(s)X(s) \quad y_{zs}(t) = \mathrm{LT}^{-1}[Y_{zs}(s)]$

2. To relate the system function to the system’s **differential equation** and **simulation block diagram.**

$$H(s) = \frac{s^{-1} + 3s^{-2}}{1 + 3s^{-1} + 2s^{-2}} = \frac{s+3}{s^2 + 3s + 2} \qquad \frac{d^2 y(t)}{dt^2} + 3\frac{dy(t)}{dt} + 2y(t) = \frac{dx(t)}{dt} + 3x(t)$$



3. To determine **the impulse response** and characteristics of **response components** by the zero-pole distribution of the system function.

4. To determine the system’s **frequency response characteristics** by the zero-pole distribution of the system function.

Zeros, Poles of System Function & Excitation, and Response Components

The **poles** of the system function and excitation jointly determine the response waveform:

$$Y(s) = X(s)H(s)$$

Poles of the system function correspond to free response components; poles of the excitation correspond to forced response components.

If **zeros and poles of the system function and excitation cancel each other**, the corresponding time component does not appear in the response.

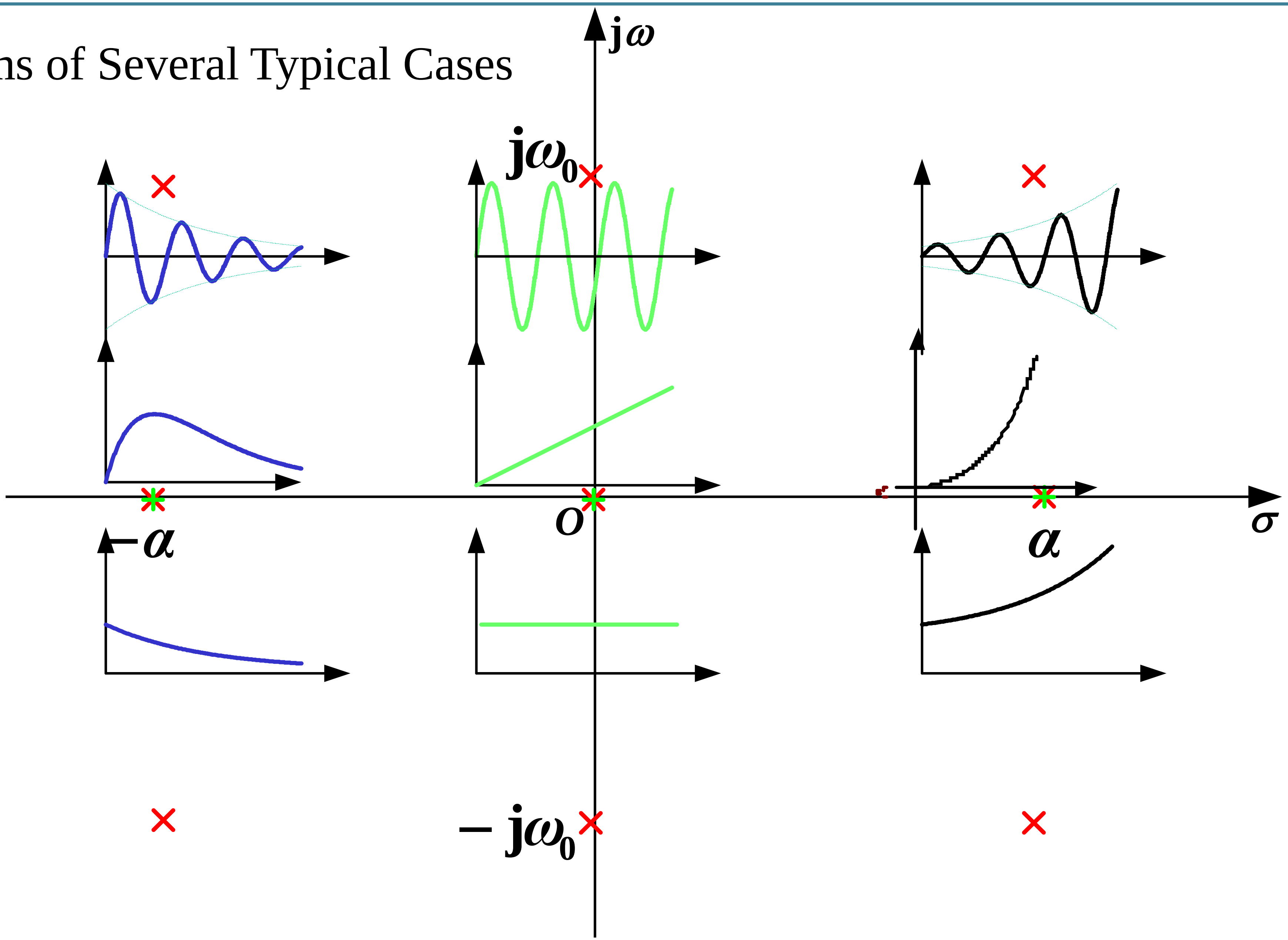
Example： Laplace transform of zero-state response

$$Y(s) = X(s)H(s) = \frac{1}{s+3} \cdot \frac{s+3}{s^2+3s+2}$$

zero-state response: $y(t) = (e^{-t} - e^{-2t})u(t)$

Only free response exists, as forced response is zero due to **zero-pole cancellation**.

Diagrams of Several Typical Cases



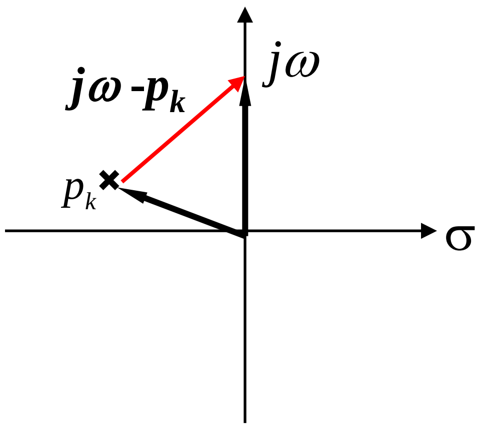
4.8 Time-domain Characteristics from System Function Zero-pole Distribution

Correspondence between Poles of ***H*(*s*)** and Waveform Characteristics of ***h*(*t*)**:

Poles on the left half-plane of the s-plane	***h*(*t*) exhibits a decaying form**
Poles in the right half-plane of the s-plane	***h*(*t*)** exhibits a growing form
Poles on the imaginary axis of the s-plane (excluding the origin)	***h*(*t*)** has constant-amplitude oscillation (first-order poles) or growing oscillation (multi-order poles)
Poles on the real axis of the s-plane (excluding the origin)	***h*(*t*)** exhibits exponential-related changes (first-order or multi-order poles)
Poles at the origin of the s-plane	***h*(*t*)** is equal to ***u*(*t*)**(first-order pole) or grows as t^n (multi-order poles)

The frequency response of a system can **generally** be expressed:

$$H(\omega)=H(s)\big|_{s=j\omega}=\left.\frac{G\prod_{k=1}^{M}(s-z_k)}{\prod_{k=1}^{N}(s-p_k)}\right|_{s=j\omega}=\frac{G\prod_{k=1}^{M}(j\omega-z_k)}{\prod_{k=1}^{N}(j\omega-p_k)}=\frac{G\prod_{k=1}^{M}B_k}{\prod_{k=1}^{N}A_k}e^{j(\sum_{k=1}^{M}\beta_k-\sum_{k=1}^{N}\theta_k)}=\left|H(\omega)\right|e^{j\varphi(\omega)}$$



$$\left|H(\omega)\right|=\frac{G\prod_{k=1}^{M}B_k}{\prod_{k=1}^{N}A_k}\qquad\qquad \varphi(\omega)=\sum_{k=1}^{M}\beta_k-\sum_{k=1}^{N}\theta_k$$

The system's magnitude-frequency response is **the ratio of the product of the magnitudes of zero vectors to that of pole vectors**.
The system's phase-frequency response is **the sum of the phase angles of zero vectors minus that of pole vectors**.

Contents of This Lecture

Learning Objectives of This Lecture

1. Understand the characteristics of zero and pole distribution of **all-pass** functions and **minimum-phase-shift** functions;
2. Skillfully use **system functions** to determine the **stability** of systems;
3. Be familiar with the correspondence between the **region of convergence** of bilateral Laplace transform and the **original function**;
4. Understand the correspondence between Laplace transform and **Fourier transform**.

4.10.1 Zero-Pole Distribution of All-pass Systems

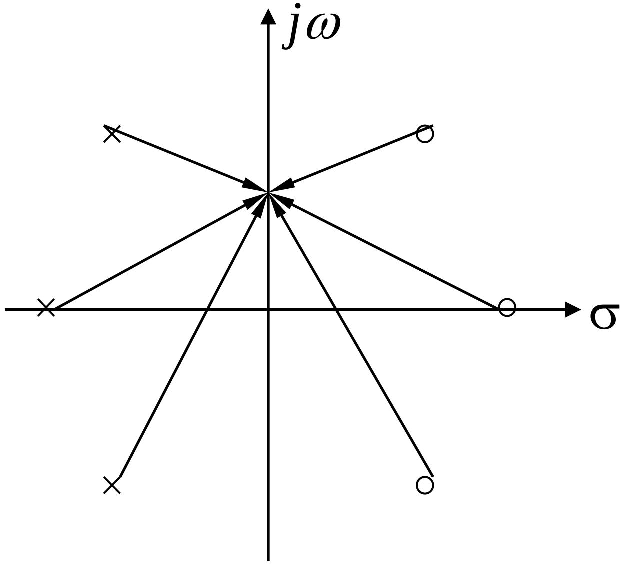
The magnitude-frequency response of an all-pass system is a constant at all frequencies.

The **phase-frequency response** of an all-pass system is **not restricted**.

The product of the magnitudes of the zero vectors and the product of the magnitudes of the pole vectors of an **all-pass system** function are equal **at all frequencies**.

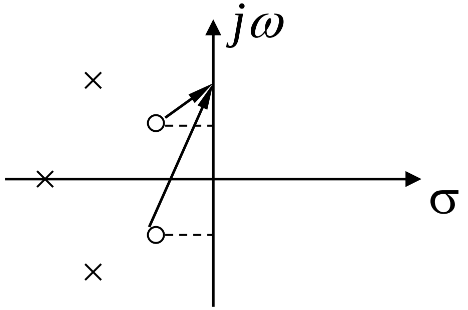
Hence, **the zeros and poles are symmetric with respect to the imaginary axis**.

$$|H(\omega)| = \frac{G\prod_{k=1}^{N} B_k}{\prod_{k=1}^{N} A_k} = G$$

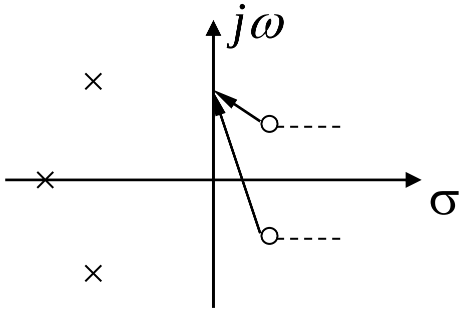


4.10.2 Zero-Pole Distribution of Minimum-phase-shift Systems

Minimum-phase-shift System: All zeros of the system function are located on the left half-plane or on the imaginary axis of the s-plane. Otherwise it is a non-minimum-phase-shift system.



Pole-Zero Plot of
Minimum-phase-shift System



Pole-Zero Plot of
Non-minimum-phase-shift System

In the above two pole-zero plots, poles are distributed the same, zeros have equal imaginary parts but opposite-signed real parts. Obviously, **their magnitude-frequency responses are identical**; at all frequencies, the phase angles of zeros in the left plot are smaller than those in the right plot, while pole phase angles are the same. Thus, **the left plot has a smaller phase shift** in absolute value.

单选题 1分

设置

Among the following system functions, which one belongs to a minimum-phase-shift system? (　)

A $H(s)=\dfrac{(s-2)^2+6}{(s+1)(s+5)^2}$

B $H(s)=\dfrac{(s-1)(s+2)}{s(s+1)^2}$

C $H(s)=\dfrac{(s+2)^2+6}{(s+1)(s+5)^2}$

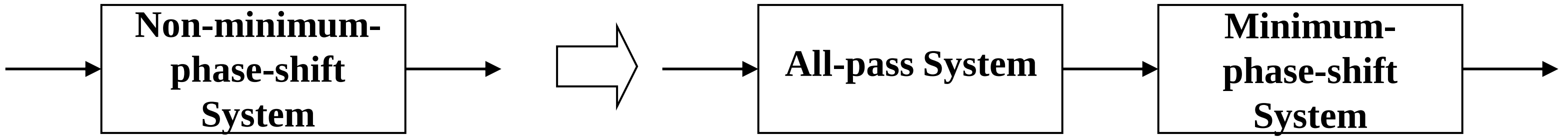
D $H(s)=\dfrac{(s-3)^2}{s^3+5s+7}$

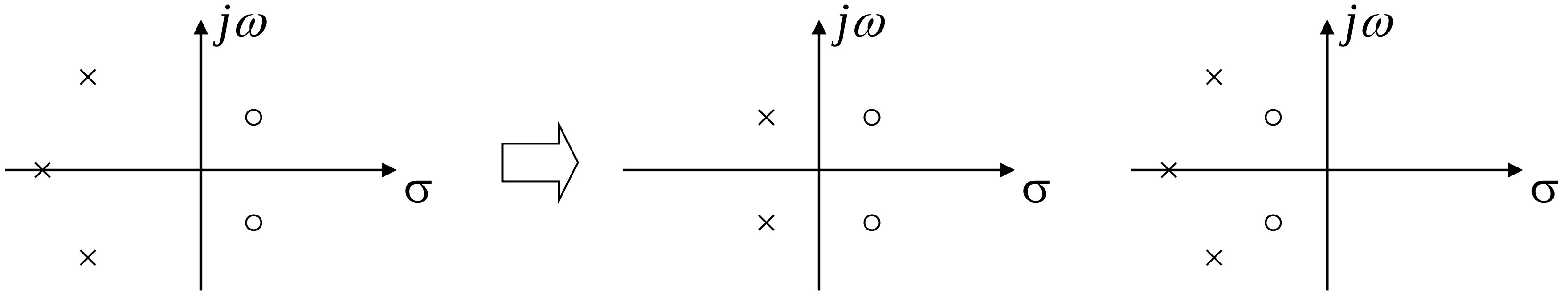
提交

4.10 Zero-Pole Locations of All-pass and Minimum-phase-shift Functions

Any non-minimum-phase-shift system can be expressed as **a cascade of an all-pass system and a minimum-phase-shift system**.

$$H(s) = H_{AP}(s) \cdot H_{\min}(s)$$





The **poles** of the all-pass system **cancel** with the **zeros** of the minimum-phase-shift system.

Example:

$$H(s) = \frac{(s-\alpha_2)^2+\Omega_2^2}{(s+\alpha_0)[(s+\alpha_1)^2+\Omega_1^2]} = \frac{(s+\alpha_2)^2+\Omega_2^2}{(s+\alpha_0)[(s+\alpha_1)^2+\Omega_1^2]} \cdot \frac{(s-\alpha_2)^2+\Omega_2^2}{(s+\alpha_2)^2+\Omega_2^2}$$

$$H_{AP}(s) = \frac{(s-\alpha_2)^2+\Omega_2^2}{(s+\alpha_2)^2+\Omega_2^2} \qquad H_{\min}(s) = \frac{(s+\alpha_2)^2+\Omega_2^2}{(s+\alpha_0)[(s+\alpha_1)^2+\Omega_1^2]}$$

4.11.1 Causality and Stability of Systems

The **stability** of a system refers to a system where **a bounded input** produces **a bounded output**. That is, if the input $|x(\cdot)| < \infty$, the output $|y(\cdot)| < \infty$, the system must be stable. Specifically**, the output cannot contain impulse functions and their derivatives.**

The stability of a linear time-invariant (LTI) system means its impulse response satisfies absolute integrability, i.e.

$$\int_{-\infty}^{\infty} |h(t)| dt < \infty$$

The **causality** of a system means that when the initial state is zero, the output does not occur before the input acts on the system. That is, if $\boldsymbol{t<t_0}$ and $\boldsymbol{x(t)=0}$, then necessarily $\boldsymbol{t<t_0}$ and $\boldsymbol{y(t)=0}$.

The causality of a linear system means its impulse response satisfies:

$$h(t) = h(t)u(t)$$

When a system is **causal and stable**, its unit impulse response should satisfy absolute integrability from 0^- to ∞:

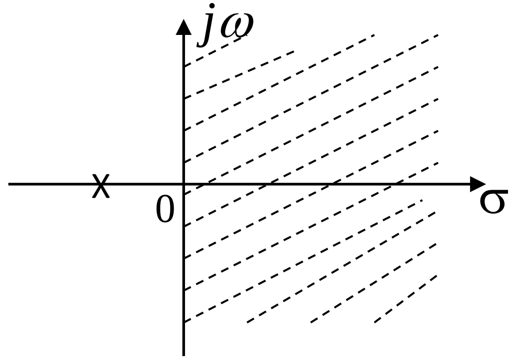
$$\boxed{\int_{0^-}^{\infty} |h(t)| dt < \infty} \qquad (1)$$

As $$H(s) = \int_{0^-}^{\infty} h(t) e^{-st} dt = \int_{0^-}^{\infty} h(t) e^{-(\sigma + j\omega)t} dt \qquad (2)$$

For (2) to be absolutely integrable,

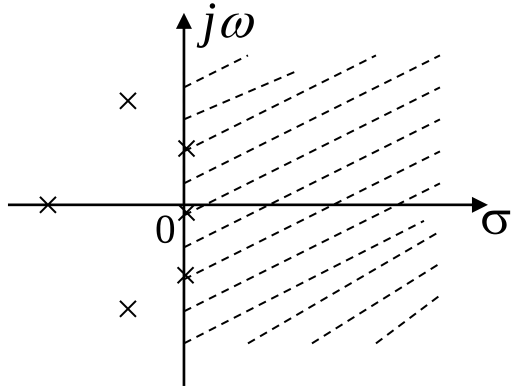
$$\int_{0^-}^{\infty} \left| h(t) e^{-\sigma t} \right| dt < \infty \qquad (3)$$

$$\left(\left| e^{-j\omega t} \right| = 1 \right)$$



By comparing (1) and (3), it can be seen that the minimal region of convergence (ROC) of the system function of a linear causal and stable system is **σ=Re{s}>0.** That is, **the poles of the system function of a causal and stable system can only be distributed on the left half-plane of the imaginary axis on the s-plane.**

What if the poles of the system function are located on the imaginary axis of the s-plane?



- **Case 1**: **a first-order pole at the origin** corresponds to the factor $\frac{1}{s}$, whose inverse transform is the unit step signal ***u*(*t*)**, which does not satisfy absolute integrability. However, **when *t*>0, its amplitude remains stable**.
- **Case 2**: **a pair of complex conjugate poles on the imaginary axis** correspond to factors like $\frac{\Omega_0}{s^2+\Omega_0^2}$ or $\frac{s}{s^2+\Omega_0^2}$, whose inverse Laplace transforms are $\sin(\Omega_0 t)$ and $\cos(\Omega_0 t)$, respectively, not satisfying absolute integrability. But **when *t*>0, their maximum amplitude remains stable**.
- **Case 3**: If the poles on the imaginary axis are of multiple orders, the corresponding time signals does not satisfy absolute integrability, and their amplitude increases when ***t*>0.**

Thus, **the poles on the imaginary axis make the system unstable**. The cases of **first-order pole and complex conjugate poles on the imaginary axis** are also referred to as **marginally stable**.

4.11 Stability of Linear Systems

A stable system also has requirements for **the number of zeros** in its system function. Let the system function be:

$$H(s)=\frac{N(s)}{D(s)}=s+\frac{A(s)}{D(s)}$$

***A*(*s*)/*D*(*s*)** is a proper rational fraction. When the bounded input ***x*(*t*)=*u*(*t*)** (with its Laplace transform being **1/s**) is applied, the Laplace transform of the output will contain 1, corresponding to the impulse function ***δ*(*t*)** in the output, which is unbounded in amplitude. It can be seen that the above system is unstable. ***h*(*t*) can contain impulse functions but not their derivatives.**

In summary, **the stability conditions in the s-domain for a linear system** are:

(1) All poles of the system function lie in the left half-plane of the s-plane's imaginary axis; real roots are negative real numbers, and complex roots have negative real parts.

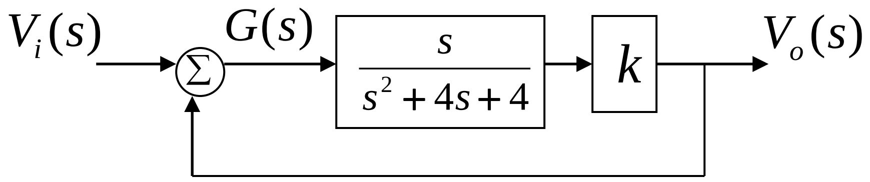
(2) The order of the system function's numerator polynomial is no higher than that of the denominator (the number of zeros does not exceed the number of poles).

Given a system with the system function $H(s)=\dfrac{s}{s^2-s-6}$, this system is ( )

A Stable

B Unstable

Example 4-30: The following figure shows a feedback system. 1) Find the system function. 2) What condition should **k** satisfy for the system to be stable? 3) When the system is marginally stable, find the impulse response of the system.



Solution: 1) $G(s)=V_i(s)+V_o(s)$

$$V_o(s)=\frac{ksG(s)}{s^2+4s+4}$$

$$\therefore H(s)=\frac{V_o(s)}{V_i(s)}=\frac{ks}{s^2+(4-k)s+4}$$

2) Poles of the system function: $p_{1,2}=\dfrac{-(4-k)\pm\sqrt{(4-k)^2-16}}{2}$

when **k>4,** the system function has poles with positive real parts, so the system is unstable. Therefore, the system is stable when **k<4.**

3) when **k=4,** the system is marginally stable. The system function is $H(s)=\dfrac{4s}{s^2+4}$.

$$\therefore h(t)=4\cos 2t u(t)$$

How to determine whether all roots of a polynomial have negative real parts?

When the polynomial is of first order or second order, the following conditions can be used to determine that **all its roots have negative real parts**:

(1) All coefficients of the denominator polynomial are non-zero, i.e., there are no missing terms;

(2) The signs of all coefficients of the denominator polynomial are the same. When the leading coefficient is 1, all other coefficients should be positive.

According to the above conditions, the stability condition of the previous example can be obtained immediately:

$$H(s)=\frac{ks}{s^2+(4-k)s+4} \qquad 4-k>0 \qquad \therefore k<4$$

Transient Response and Steady-State Response of Linear Stable Systems

- **Transient Response**: Under the action of excitation (**step signal or causal periodic signal**), it is a component that appears instantaneously in the total response and gradually disappears as time increases.
- **Steady-State Response:** when $t \to \infty$, it is the component retained in the total response, usually composed of **step functions** and **periodic functions**.
- The poles of the system function of a stable system **lie on the left half-plane of the s-plane,** and the free response function shows an attenuating trend. **Free response = Transient response.**
- If the poles of the excitation **lie on the imaginary axis or in the right half-plane of the s-plane, Forced response = Steady-state response.**

Example: Excitation signal ***x*(*t*)=cos(2*t*)*u*(*t*),** the poles of the system function are -6 and -1, and the total response is

$$r(t) = \underbrace{-\frac{6}{25}e^{-t} + \frac{9}{50}e^{-6t}}_{\substack{\textbf{Free response} \\ \textbf{Transient Response}}} + \underbrace{\frac{21}{50}\sin(2t) + \frac{3}{50}\cos(2t)}_{\substack{\textbf{Forced response} \\ \textbf{Steady-state response}}} \quad (t \geq 0)$$

Bilateral Laplace Transform:

$$F_B(s)=\int_{-\infty}^{\infty} f(t)e^{-st}dt=\int_{-\infty}^{\infty} f(t)e^{-\sigma t}e^{-\mathrm{j}\omega t}dt$$

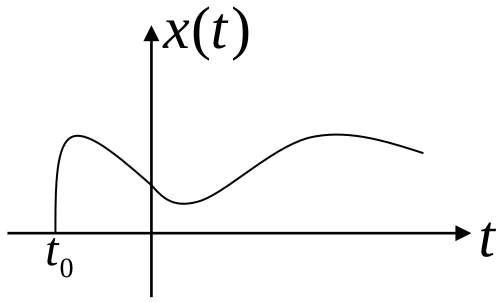
For the attenuation factor $\boldsymbol{\sigma}$**,** the situation when $\boldsymbol{t}$**>0** is exactly opposite to that when $\boldsymbol{t}$**<0.** Therefore, **the integral result of the bilateral Laplace transform does not necessarily exist**, which is different from the unilateral Laplace transform. We need to discuss the existence of the bilateral Laplace transform.

4.12.1 Region of Convergence (ROC) of Bilateral Laplace Transform

The **ROC** of a signal's Laplace transform is related to the form of the signal itself. According to the definition of the Laplace transform, the existence of the Laplace transform for a general signal should satisfy:

$$\lim_{t\to\infty} x(t)e^{-\sigma t}=0 \quad \& \quad \lim_{t\to-\infty} x(t)e^{-\sigma t}=0$$

4.12.2 Bilateral Laplace Transform of Right-Sided Signals



When a signal is a **right-sided signal**, i.e. ***x*(*t*)=0** for ***t*<*t*$_0$,** (**causal signals** are **right-sided signals**), its Laplace transform is:

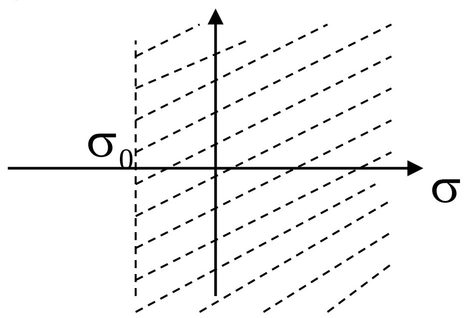
$$X(s)=\int_{t_0}^{\infty} x(t)e^{-st}dt=\int_{t_0}^{\infty} x(t)e^{-\sigma t}e^{-j\omega t}dt$$

if $\boldsymbol{\sigma_0}$ makes the above integral absolutely integrable, i.e., $\int_{t_0}^{\infty}\left|x(t)e^{-\sigma_0 t}e^{-j\omega t}\right|dt=\int_{t_0}^{\infty}\left|x(t)e^{-\sigma_0 t}\right|dt<\infty$

Let $\sigma=\sigma_0+\Delta$, given $\lim_{t\to\infty} x(t)e^{-\sigma_0 t}=0$, for $x(t)e^{-\sigma t}$ to be absolutely integrable,

we have: $\lim_{t\to\infty} x(t)e^{-\sigma t}=\lim_{t\to\infty} x(t)e^{-\sigma_0 t}e^{-\Delta t}=0$,

which requires $\Delta>0$, i.e. $\sigma>\sigma_0$.



Thus, the region of convergence (ROC) is $\mathrm{Re}\{s\}=\sigma>\sigma_0$

单选题 1分

设置

Given the signal $x(t) = e^{-t}u(t) - e^{-2t}u(t)$， its bilateral Laplace transform and ROC are ( )

A $X(s) = \dfrac{-1}{(s-1)(s-2)}, \sigma > -2$

B $X(s) = \dfrac{1}{(s+1)(s+2)}, \sigma > -1$

C $X(s) = \dfrac{-1}{(s-1)(s-2)}, \sigma > -1$

D $X(s) = \dfrac{1}{(s+1)(s+2)}, \sigma > -2$

提交

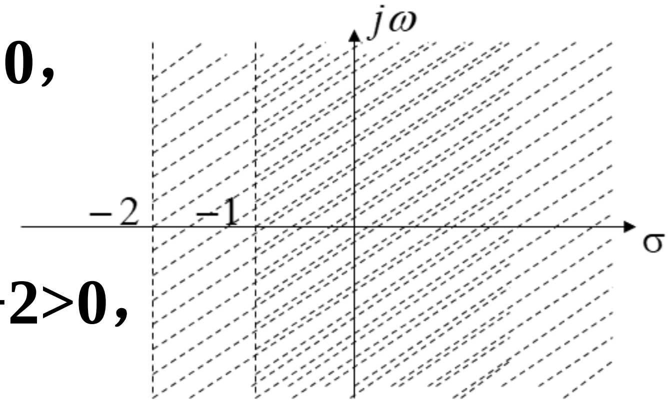
Example 4-31: Find the bilateral Laplace transform of the following causal signal and specify its ROC.

$$x(t)=e^{-t}u(t)-e^{-2t}u(t)$$

Solution: $X(s)=\int_{-\infty}^{\infty} x(t)e^{-st}dt = \int_{0}^{\infty} e^{-t}e^{-st}dt-\int_{0}^{\infty} e^{-2t}e^{-st}dt$

$$=\int_{0}^{\infty} e^{-(s+1)t}dt-\int_{0}^{\infty} e^{-(s+2)t}dt=\int_{0}^{\infty} e^{-(\sigma+1)t}e^{-j\omega t}dt-\int_{0}^{\infty} e^{-(\sigma+2)t}e^{-j\omega t}dt$$

For the first term, the integral is integrable when **Re{s+1}=σ+1>0,** and $\frac{1}{s+1}$ has an ROC of $\mathrm{Re}\{s\}=\sigma>-1$

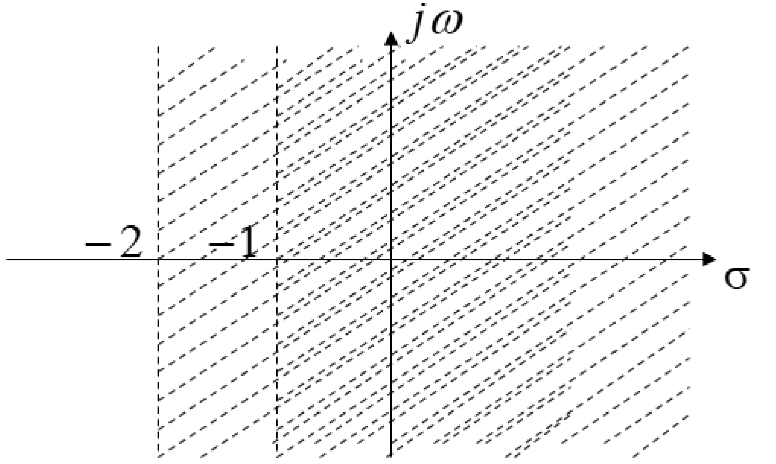


For the second term, the integral is integrable when **Re{s+2}=σ+2>0,** and $\frac{1}{s+2}$ has an ROC of $\mathrm{Re}\{s\}=\sigma>-2$

Thus, the Laplace transform of the entire function is:

$$X(s)=\frac{1}{s+1}-\frac{1}{s+2}=\frac{1}{(s+1)(s+2)}$$

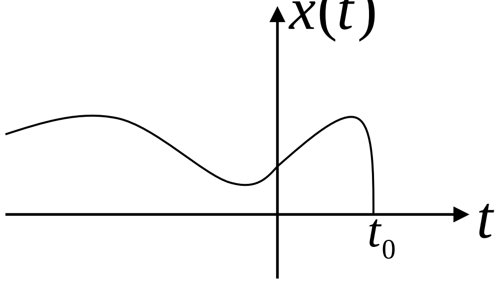
The ROC is $\mathrm{Re}\{s\}=\sigma>-1$



That is, the ROC of the signal's Laplace transform is **the common region of the ROCs of the two parts**. **The ROC of a right-sided signal is to the right of its rightmost pole**.

4.12.3 Bilateral Laplace Transform of Left-Sided Signals (Anti-Causal Signals)

When a signal is a **left-sided signal**, i.e. $\boldsymbol{x(t)=0}$ for $\boldsymbol{t>t_0}$ (it is an **anti-causal signal** if $\boldsymbol{t_0=0}$**),** its Laplace transform is:

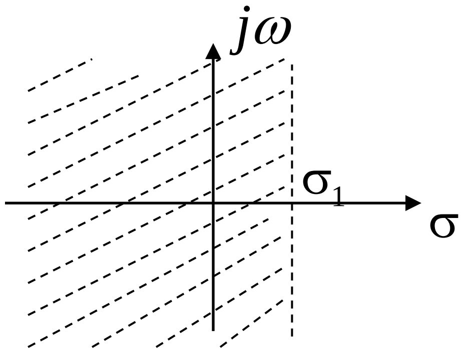


$$X(s)=\int_{-\infty}^{t_0} x(t)e^{-st}dt=\int_{-\infty}^{t_0} x(t)e^{-\sigma t}e^{-j\omega t}dt$$

if $\boldsymbol{\sigma_1}$ makes the above integral absolutely integrable, i.e.,

$$X(s)=X(\sigma_1+j\omega)=\int_{-\infty}^{t_0} x(t)e^{-\sigma_1 t}e^{-j\omega t}dt$$

exists, then the ROC is $\mathrm{Re}\{s\}=\sigma<\sigma_1$



Proof: Let $\sigma=\sigma_1+\Delta$. Given $\lim\limits_{t\to-\infty} x(t)e^{-\sigma_1 t}=0$,

for $x(t)e^{-\sigma t}$ to be absolutely integrable, we have:

$$\lim_{t\to-\infty} x(t)e^{-\sigma t}=\lim_{t\to-\infty} x(t)e^{-\sigma_1 t}e^{-\Delta t}=0,\ \text{ which requires } \Delta<0,\ \text{ i.e. } \sigma<\sigma_1.$$

Example 4-32: Given the anti-causal signal $\boldsymbol{x(t)=-e^{-\alpha t}u(-t)}$, find its bilateral Laplace transform $\boldsymbol{X(s)}$.

Solution: By definition

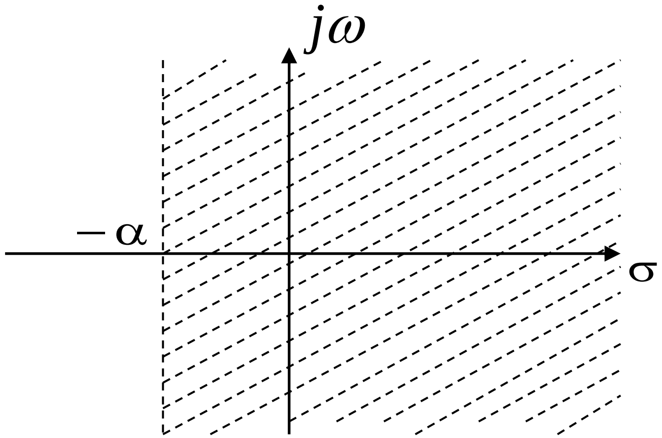
$$X(s)=\int_{-\infty}^{\infty}x(t)e^{-st}dt=-\int_{-\infty}^{\infty}e^{-\alpha t}u(-t)e^{-st}dt=-\int_{-\infty}^{0}e^{-\alpha t}e^{-st}dt$$

$$=-\int_{-\infty}^{0}e^{-(s+\alpha)t}dt$$

when $\mathrm{Re}\{s+\alpha\}=\sigma+\alpha<0$， $\sigma<-\alpha$， the integral exists.

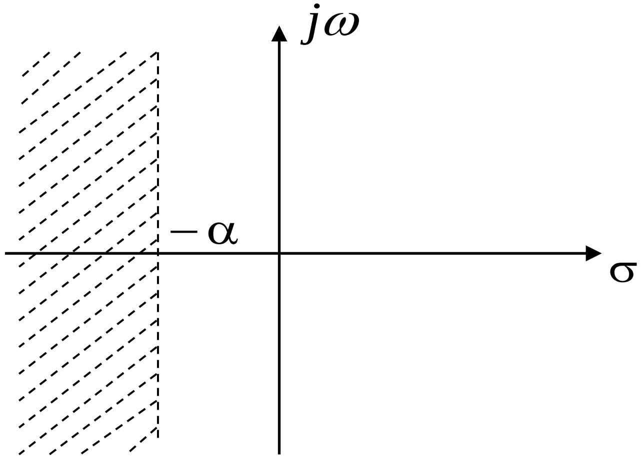
$$X(s)=\frac{1}{(s+\alpha)}e^{-(s+\alpha)t}\Big|_{-\infty}^{0}=\frac{1}{(s+\alpha)},\quad \sigma<-\alpha$$

4.12 Bilateral Laplace Transform



Bilateral Laplace Transform of $e^{-\alpha t}u(t)$

$$e^{-\alpha t}u(t) \xleftrightarrow{LT} 1/(s+\alpha), \quad \mathrm{Re}\{s\} > -\alpha$$



Bilateral Laplace Transform of $-e^{-\alpha t}u(-t)$

$$-e^{-\alpha t}u(-t) \xleftrightarrow{LT} 1/(s+\alpha), \quad \mathrm{Re}\{s\} < -\alpha$$

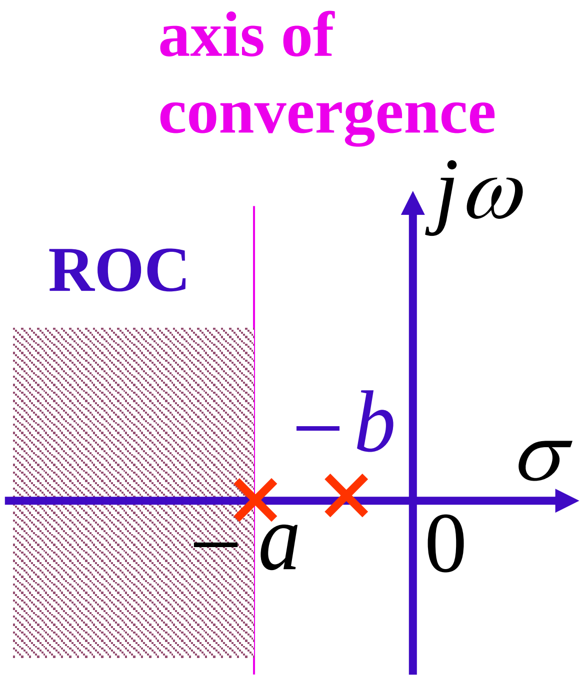
The ROCs of the bilateral Laplace transforms of the above two signals are both half-open planes on the s-plane. **-α** is called their abscissa of convergence, and **σ=Re{s}=-α** is called their **axis of convergence** or **boundary of the ROC**.

The same bilateral Laplace transform expression can represent different time signals due to different ROCs. Therefore, when representing the original signal by its bilateral Laplace transform, **the ROC must be specified; otherwise, the original signal cannot be determined**.

4.12 Bilateral Laplace Transform

Given the signal: $f(t) = (e^{-at} + e^{-bt})u(-t) \quad (a > b > 0)$

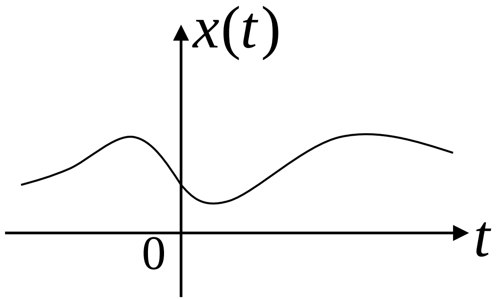
The image function is: $F(s) = \dfrac{-1}{s+a} + \dfrac{-1}{s+b} \quad (\sigma < -a)$



The ROC of a left-sided signal lies to the left of its leftmost pole.

All poles of the image function are located to the right of the ROC.

4.12.4 Bilateral Laplace Transform of Bilateral Signals

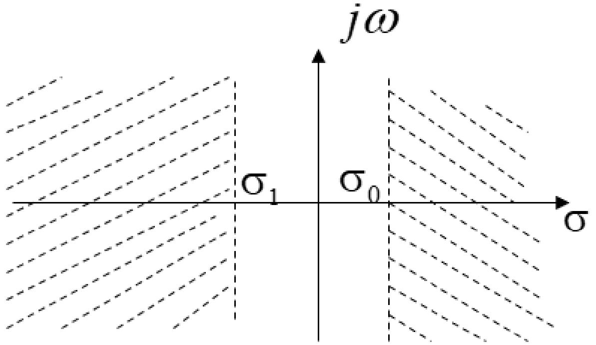


If a signal is a **bilateral signal**, its bilateral Laplace transform is:

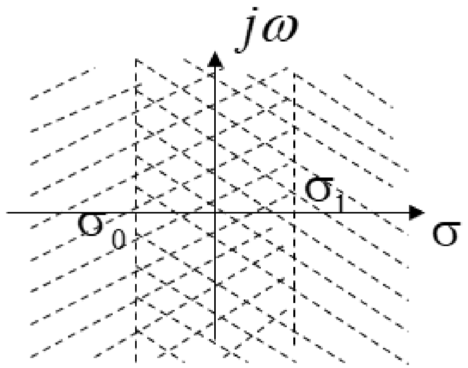
$$X(s)=\int_{-\infty}^{\infty}x(t)e^{-st}dt=\int_{-\infty}^{0}x(t)e^{-\sigma t}e^{-j\omega t}dt+\int_{0}^{\infty}x(t)e^{-\sigma t}e^{-j\omega t}dt$$

If $\boldsymbol{\sigma_1}$ makes the first term of the above expression absolutely integrable, and $\boldsymbol{\sigma_0}$ makes the second term absolutely integrable, then:

$$X(s)=\int_{-\infty}^{0}x(t)e^{-\sigma_1 t}e^{-j\omega t}dt+\int_{0}^{\infty}x(t)e^{-\sigma_0 t}e^{-j\omega t}dt$$



if $\sigma_0>\sigma_1$， $X(s)$ does not exist



if $\sigma_0<\sigma_1$， $X(s)$ exists，
the ROC is $\sigma_0<\sigma<\sigma_1$

单选题 1分

设置

Given $f(t)=u(t)+e^{t}u(-t)$， its bilateral Laplace Transform is (   )

A $F(s)=\dfrac{1}{s}-\dfrac{1}{s-1}$ ， $\sigma>0$

B $F(s)=\dfrac{1}{s}+\dfrac{1}{s+1}$ ， $0<\sigma<1$

C $F(s)=\dfrac{1}{s}-\dfrac{1}{s-1}$ ， $0<\sigma<1$

D $F(s)=\dfrac{1}{s}+\dfrac{1}{s+1}$ ， $\sigma>-1$

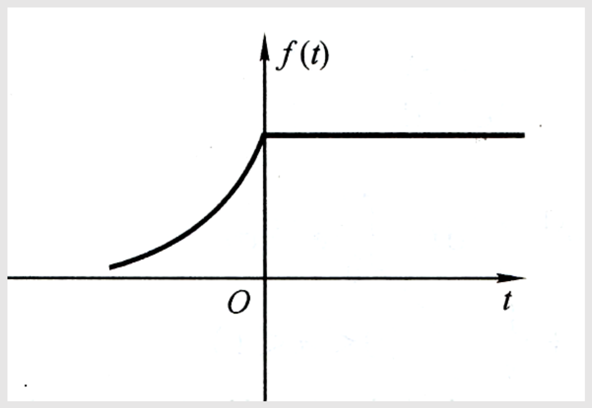
提交

4.12 Bilateral Laplace Transform

Example 4-33: Given $f(t)=u(t)+e^{t}u(-t)$, find its bilateral Laplace Transform.

Solution:

$f(t)$ $\qquad$ $f(t)u(t)$ $\qquad$ $f(t)u(-t)$



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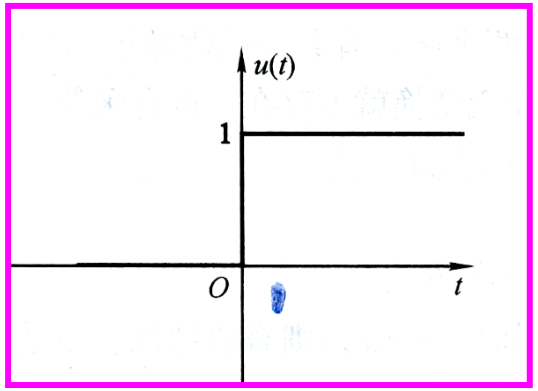

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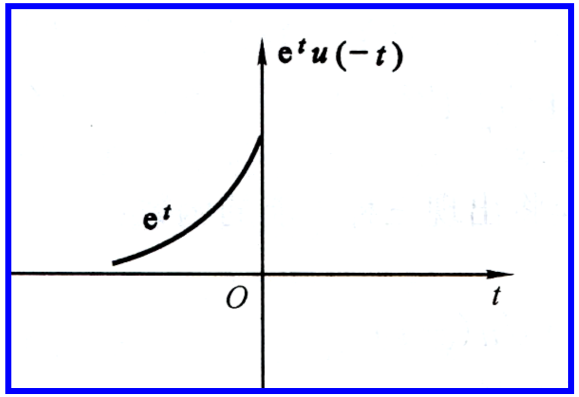
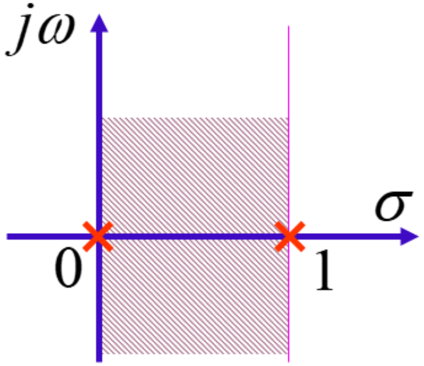


$$F(s)=\int_{-\infty}^{\infty}f(t)e^{-st}dt=\int_{-\infty}^{0}e^{t}e^{-st}dt+\int_{0}^{\infty}e^{-st}dt=\int_{-\infty}^{0}e^{(1-\sigma)t}e^{-j\omega t}dt+\int_{0}^{\infty}e^{-\sigma t}e^{-j\omega t}dt$$

To make the two terms in the above expression absolutely integrable, $1-\sigma>0$ and $\sigma>0$, so $0<\sigma<1$.

$$F(s)=\frac{1}{s}+\frac{1}{1-s}\qquad(0<\sigma<1)$$

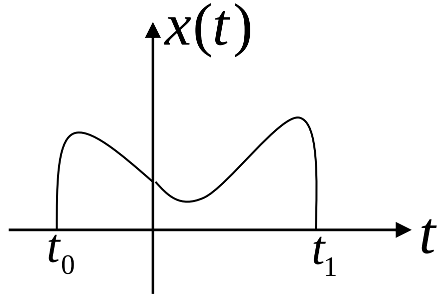


ROC

4.12.5 Bilateral Laplace Transform of Time-Limited Signals

If a signal is a **time-limited signal**, i.e. $t_0>t>t_1$, $x(t)=0$, its Laplace transform is defined as follows:

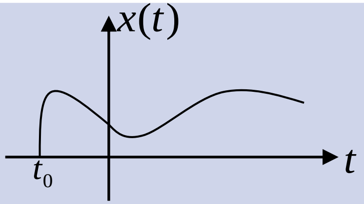
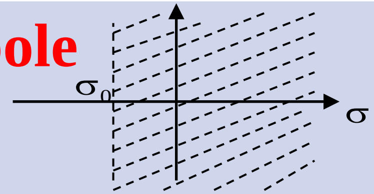
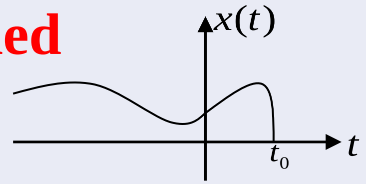
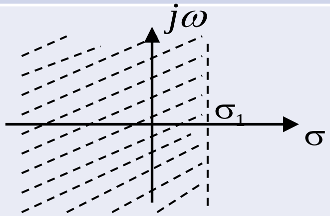
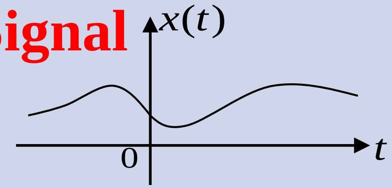
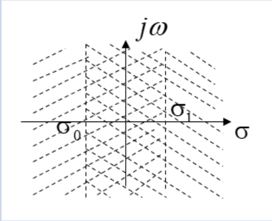
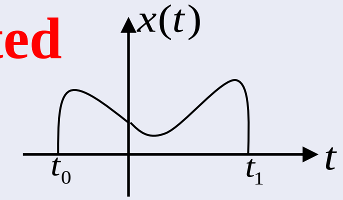
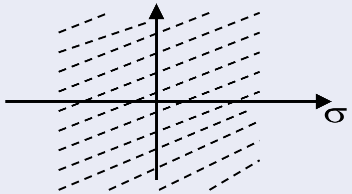
$$X(s)=\int_{-\infty}^{\infty} x(t)e^{-st}dt=\int_{t_0}^{t_1} x(t)e^{-st}dt=\int_{t_0}^{t_1} x(t)e^{-\sigma t}e^{-j\omega t}dt$$



At this time, if $x(t)$ is absolutely integrable, $x(t)e^{-\sigma t}$ is always absolutely integrable. Therefore, its bilateral Laplace transform exists over the entire s-plane, **i.e.**
the ROC is the entire s-plane.

4.12.6 Characteristics of the ROC for Bilateral Laplace Transform

When considering the existence of the ROC for the bilateral Laplace transform, **the ROC must be specified.**

Signal Category	ROC of Bilateral Laplace Transform
Right-Sided Signal 	To **the right of the rightmost pole** of the image function 
Left-Sided Signal 	To **the left of the leftmost pole** of the image function 
Bilateral Signal 	**Band-shaped region** between two poles (if the bilateral Laplace transform exists) 
Time-Limited Signal 	**Entire s-plane** 

4.12.7 Inverse Bilateral Laplace Transform

It can be solved based on the ROC and the poles:

(1) Perform partial fraction decomposition on the image function;

(2) Take the inverse transform of the image function according to the poles;

(3) **Poles to the left of the ROC correspond to right-sided signals, and poles to the right of the ROC correspond to left-sided signals**.

F(s)	**ROC**	**f(t)**
$1/s$	$\sigma>0$	$u(t)$
$1/s$	$\sigma<0$	$-u(-t)$
$1/(s+\alpha)$	$\sigma>-\alpha$	$e^{-\alpha t}u(t)$
$1/(s+\alpha)$	$\sigma<-\alpha$	$-e^{-\alpha t}u(-t)$

Example 4-34: Given $F(s)=\frac{1}{s}+\frac{1}{s+\alpha}$ $(\alpha>0)$, find all possible inverse bilateral Laplace transforms.

Solution:

According to the poles of the image function, 0 and -α, there are **three possible cases** for its ROC, and different ROCs yield different time-domain signals when taking the inverse transform.

$$\sigma>0 \qquad f(t)=u(t)+e^{-\alpha t}u(t)$$

$$-\alpha<\sigma<0 \qquad f(t)=-u(-t)+e^{-\alpha t}u(t)$$

$$\sigma<-\alpha \qquad f(t)=-u(-t)-e^{-\alpha t}u(-t)$$

4.13 Relationship between Laplace Transform and Fourier Transform

The Laplace transform is an extension of the Fourier transform from the real frequency **ω** to the complex frequency ***s*=*σ*+*j*ω**， and the Fourier transform is a special case of the Laplace transform on the imaginary axis of the ***s***-plane.

$$X(s) = \mathcal{L}\{x(t)\} = \mathcal{F}\left\{x(t)e^{-\sigma t}\right\} = X(\sigma + j\omega)$$

For many signals, replacing ***j*ω** with **s** in their Fourier transform gives their Laplace transform, and vice versa. For example, the one-sided exponential decay signal**:**

$$e^{-\alpha t}u(t) \xleftrightarrow{FT} \frac{1}{j\omega+\alpha} \qquad \therefore e^{-\alpha t}u(t) \xleftrightarrow{LT} \frac{1}{s+\alpha}$$

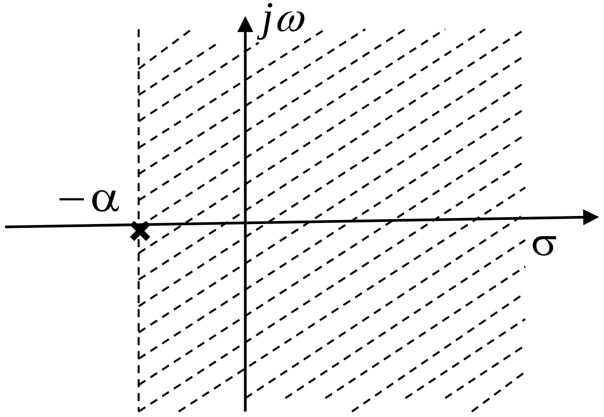
However, there are also **exceptions**. For example, the unit step signal:

$$u(t) \xleftrightarrow{FT} \pi\delta(\omega) + \frac{1}{j\omega} \qquad u(t) \xleftrightarrow{LT} \frac{1}{s}$$

4.13 Relationship between Laplace Transform and Fourier Transform

If the ROC of a signal's Laplace transform includes the imaginary axis, its Fourier transform can be obtained by substituting $s=j\omega$ into the Laplace transform.

For example, the poles of a one-sided exponential decay signal lie on the negative real axis.



ROC of Laplace Transform for $e^{-\alpha t}u(t)$

Example of a multiple pole on the negative real axis:

$$te^{-\alpha t}u(t) \xleftrightarrow{FT} \frac{1}{(j\omega+\alpha)^2}$$

$$te^{-\alpha t}u(t) \xleftrightarrow{LT} \frac{1}{(s+\alpha)^2}$$

Example of conjugate complex poles with negative real:

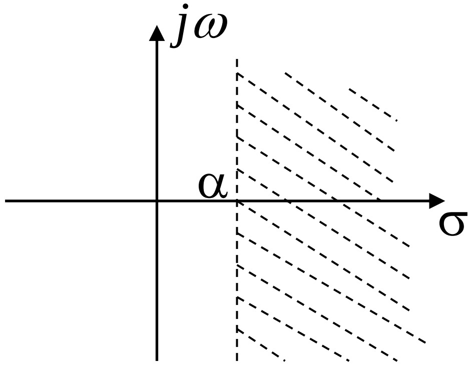
$$e^{-\alpha t}\cos(\Omega_0 t)u(t) \xleftrightarrow{FT} \frac{j\omega+\alpha}{(j\omega+\alpha)^2+\Omega_0^2}$$

$$e^{-\alpha t}\cos(\Omega_0 t)u(t) \xleftrightarrow{LT} \frac{s+\alpha}{(s+\alpha)^2+\Omega_0^2}$$

4.13 Relationship between Laplace Transform and Fourier Transform

If the ROC of the Laplace transform does not include the imaginary axis, i.e., the poles of the signal's Laplace transform lie on the right-half plane of the s-plane , **its Fourier transform does not exist**.

For example, a one-sided exponentially growing signal has its poles on the positive real axis.



$$e^{\alpha t}u(t) \xleftrightarrow{LT} \frac{1}{s-\alpha}$$

ROC of Laplace Transform for $e^{\alpha t}u(t)$

Since the signal grows exponentially and does not satisfy the condition of absolute integrability, its Fourier transform does not exist.

4.13 Relationship between Laplace Transform and Fourier Transform

When the Laplace transform of a signal **has poles located on the imaginary axis of the s-plane**, **we cannot simply substitute *jω* for *s* to obtain its Fourier transform.** For example, the unit step signal ***u*(*t*):**

$$u(t) \xleftrightarrow{LT} \frac{1}{s} \qquad\qquad u(t) \xleftrightarrow{FT} \pi\delta(\omega) + \frac{1}{j\omega}$$

Suppose the Laplace transform ***X*(*s*)** of a signal ***x*(*t*)** has a ***k***-order pole on the imaginary axis: $\boldsymbol{j\Omega_0}$

$$X(s) = X_1(s) + \frac{A_0}{(s - j\Omega_0)^k}$$

Poles are located to the left of the imaginary axis

It can be proved that its corresponding Fourier transform is:

$$X(\omega) = X(s)\big|_{s=j\omega} + \frac{\pi A_0 j^{k-1}}{(k-1)!}\delta^{(k-1)}(\omega - \Omega_0)$$

Homework

- Repeat the examples in the lecture notes.
- 6.43(a) and 6.50(b) in the textbook.